

AUGUST 12-14TH 2025

CLEAN ENERGY ROADSHOW

Learn more about:
Propane, CNG, EV, and H2
for fleets



CleanEnergyRoadshow.com



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Ted Arnold
Kia Georgia



Kia Georgia, Inc.

Kia Georgia builds 5 different models on one production line
3 Internal Combustion Engine Vehicles and 2 All Electric

November 16, 2024 – Celebrated 15 years

Total Site:	2,200 acres	Total Investment
Developed:	700 acres	\$3.2 Billion
Under Rooftop:	40 acres	

Full Production Capacity

350,000

vehicles per year



1 New Car every
51 seconds

Employment

3,200

Kia Georgia Team Members

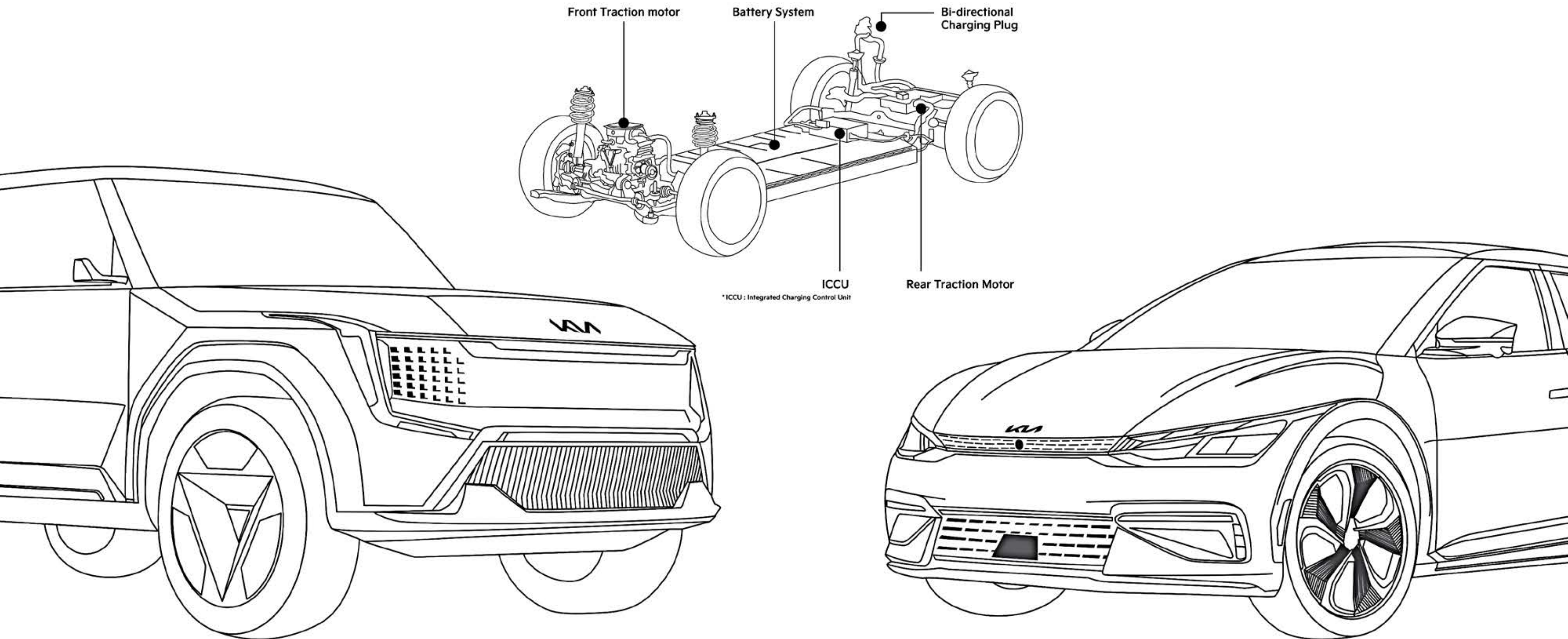
Operation

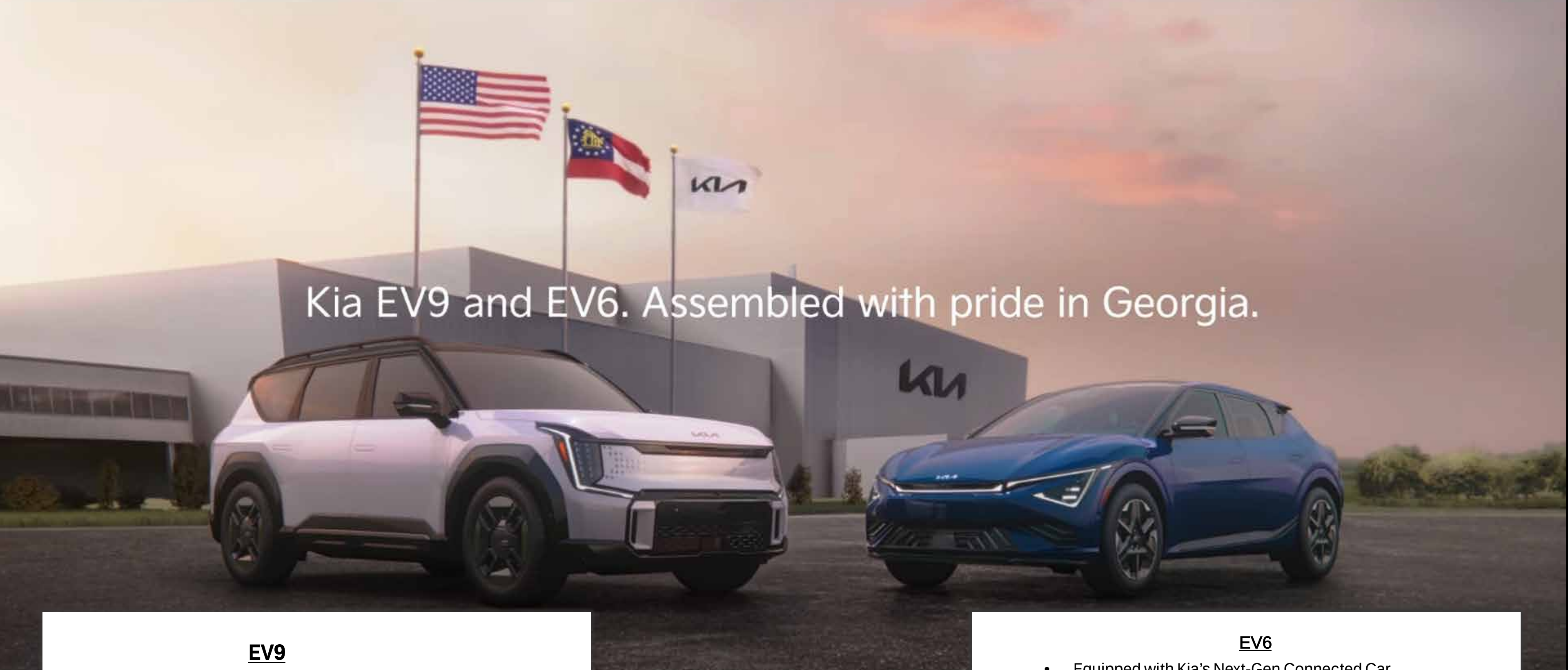
3 Shifts

5 days per week

Plan S

“S” means ‘preemptive shift’ from being a traditional automotive OEM,
into one focused on EVs and new mobility solutions



A promotional image for Kia EV9 and EV6. The background shows a modern industrial building, likely a factory, with three flags flying in front: the United States flag, the Malaysian flag, and a white flag with the Kia logo. Two cars are parked in the foreground: a white Kia EV9 SUV on the left and a blue Kia EV6 sedan on the right. The sky is a soft, hazy orange and pink, suggesting dawn or dusk. The text "Kia EV9 and EV6. Assembled with pride in Georgia." is overlaid in white on the upper part of the image.

Kia EV9 and EV6. Assembled with pride in Georgia.

EV9

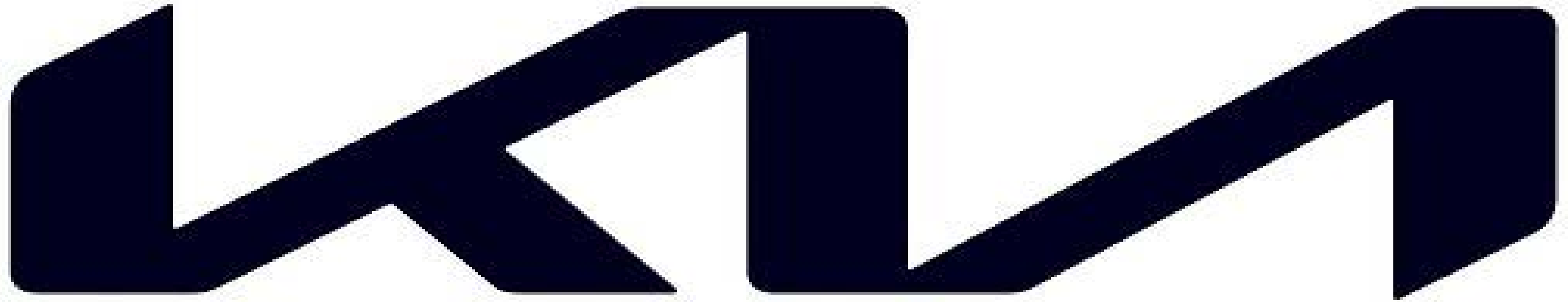
- First EV built in Georgia, May 2024
- Kia's first all-electric three-row SUV
- Targeted range 300 miles
- 10% - 80% charge in 20 minutes with 350kW charger
- Up to 379 HP with the AWD GT model
- 0-60 in 5.0 Seconds

EV6

- Equipped with Kia's Next-Gen Connected Car Navigation Cockpit
- Available with an 84.0 kWh battery pack
- Up to 319-mile range
- 10% - 80% charge in 20 minutes with 350kW charger
- Up to 601 HP with the AWD GT model
- 0-60 in 3.6 Seconds
- Earned Over 30 global awards

- Kia Georgia Solar Canopy Project





Movement that inspires

To schedule your factory tour, visit: tours@kiageorgia.com
Or call 706-902-7771



Rivian and Georgia

August 2025

Confidential. For internal use only. Do Not Distribute.

Rivian at a glance / 2025



Founded in 2009,
based in Irvine, CA

#1

Voted most loved automotive
brand amongst vehicle owners
(Consumer Reports, 2024)

130,000+

vehicles built in Illinois since
start of production in
September 2021

100%

of Rivian's charging network is
powered by renewable energy

\$4.4B

revenue generated
in 2023

14,000+

employees



Our products

Award-winning electric vehicles that are safe, reliable and fun to drive with an EPA estimated range of up to 420 miles of range.

R1T

A truck built for
any road and any task

Starting at \$69,900



R1S

A 7-seat SUV for
big adventures

Starting at \$75,900



Delivery 500 / 700

A van in two sizes for fleets,
making deliveries greener

Starting at \$79,900



Future products

Soon, the adventurous vehicles we create will be more accessible than ever.

R2

Affordable and spacious
5-seat SUV, coming 2026

Starting at \$45,000 in 2026



R3

A crossover that combines
space and nimbleness



R3X

The performance version of
the upcoming R3 crossover



Our Next Chapter: Georgia Plant

- Georgia will be home to Rivian's second flagship manufacturing plant.
- This plant will be strategically located on I-20, east of Atlanta, in Morgan and Walton Counties and will produce up to 400,000 all-electric vehicles each year.
- The state-of-the-art facility will generate at least 7,500 direct jobs, stimulating economic activity and supporting nearby communities and businesses.
- The plant is being designed as a "fossil-free plant."
- SSN construction resuming in Q3 2025 and will be operational in 2028.



Our Next Chapter: East Coast HQ

- Atlanta will be the home to Rivian's East Coast Headquarters
- Brand new location right on the Atlanta Beltline's Eastside Trail
- Currently undergoing interior build-out; opening early 2026
- Between 500-550 employees when fully operational





**KEEP
THE WORLD
ADVENTUROUS
FOREVER.**



BLUE BIRD®

ELECTRIC BUS

Overview

Best Fit Today For EV Benefits

Blue Bird Corporation/Westinghouse 1st EV School Bus (1994)

Current EV Platform launched 8 years ago

EPA Clean School Bus Program has provided education

The buses are deployed in Urban, Suburban and Rural areas

They are running in all geographies and climates

Long routes and trip buses can be managed, but are a challenge



By The Numbers

We continue to deploy and are currently in 41 US States and 4 Canadian Provinces

Most of the traditional features are available on EVs

Produced over 2500 Blue Bird EV School Buses (2000+ in the US)

Most of these buses are V2G capable providing additional value

Blue Bird holds nearly 40% of the EV School Bus Deployments



OVERVIEW | AT A GLANCE | KEY PERFORMANCE

Vision and All American Electric

- Charge Time: 3-8 hours
 - Typical 11.5kW to 30kW Charging
- Range: up to 150 mi
 - 194kW Battery Options
- Capacity Up to:
 - Vision 77 Passengers
 - All American 84 Passengers
- GVWR up to:
 - 33,000 lbs. Vision
 - 36,200 lbs. All American
- Propulsion System: Cummins PowerDrive 7000
- Motor: DANA TM4 SUMO (MN#HV3000-6P)
 - 315 HP & 2400 ft-lb of torque
 - Single Speed direct to axle (no transmission)
 - Same 5.29 axle found in ICE bus



Blue Bird Vision and All American EV



Both Type C and Type D are Available

- The Body Remains the Same
 - Similar Features to Gas/Propane/Diesel (ICE)
 - Allows for easy replacement of existing buses
 - “Plug and Play” for a Route Bus Application
- Driver Orientation
 - Assists in Familiarization
 - Simplicity of the Differences
 - Provides initial Confidence
 - Learning Charging and Range
- Key EV Benefits
 - Improved Air Quality
 - Lower Noise – Calmer and Quieter Passenger Experience
 - Reduced Maintenance Items – Lower Operational Costs
 - Better Driver Experience
 - Funding Readily Available





INTRODUCTION TO SAMSUNG LITHIUM BATTERIES



SAMSUNG LITHIUM POWERTRAIN



- ü Powered by *Samsung SDI*
- ü Nearly 10 Years of Collaboration
- ü First Golf Cart Manufacturer to Offer Lithium Drivetrain
- ü Names You Can Trust



THE NUMBERS IT TAKES TO BE #1

88 MILLION
FIELD HOURS OF PROVEN
PERFORMANCE

230,000
CARS AND COUNTING

3,750+
COURSES



#1

MOST EFFICIENT
ELECTRIC
GOLF CAR

279

LBS
LIGHTER THAN
LEAD-ACID

20 MILES

PER BATTERY PACK ON
A SINGLE CHARGE

2X

FASTER
CHARGE THAN
LEAD-ACID

ZERO

MAINTENANCE
BATTERY

5YR

UNLIMITED AMP HOUR
WARRANTY



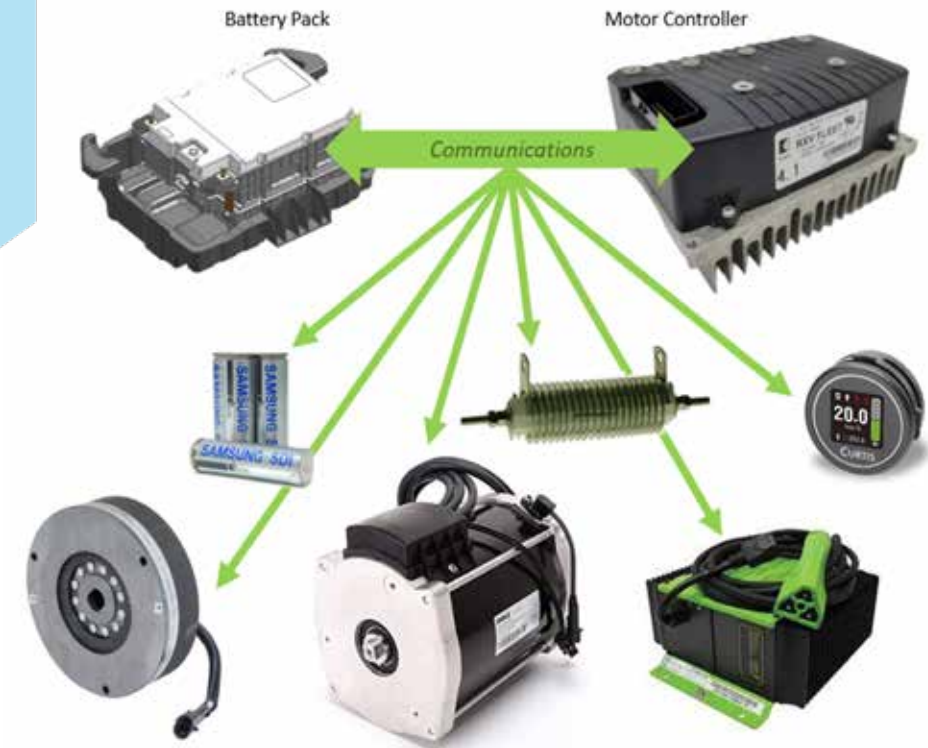
CUSHMAN



INTEGRATED LITHIUM TECHNOLOGY

ELITE
LITHIUM

- ü A safe and reliable lithium vehicle should be **INTEGRATED** into the vehicle's drivetrain system and have **REDUNDANCY** on critical functions.
- ü E-Z-GO is the only manufacturer currently with this offering. Our Battery Management System Is Integrated Directly into the Lithium Battery Pack.
- ü Integration = Reliability and Safety



Maintenance Free Batteries



Energy Efficient

59% MORE EFFICIENT VS. CLUB CAR

52% MORE EFFICIENT VS. YAMAHA

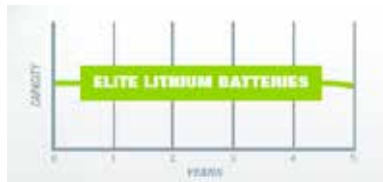
5 Year Warranty



Turf Protection



Proven Consistent Performance



Intuitive Operation



Environmentally Friendly



CUSHMAN

EZGO

DRIVING THE FUTURE OF TRANSPORTATION

TEXTRON
SPECIALIZED VEHICLES

CUSHMAN



EZGO



JACOBSEN



pace
TECHNOLOGY



TEXTRON GSE





TOGETHER FOR
CHANGE

“It’s good for business and it’s good for the world.”

[EZGO.COM/GREEN](https://ezgo.com/green)

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COX AUTOMOTIVE

GA Clean Energy Roadshow

August 2025

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Economic and Industry Insights

Autotrader  CentralDispatch  Dealer.com  Dealertrack  EV Battery Solutions  FleetNet America  Fleet Services  Kelley Blue Book  Manheim  NextGear Capital  vAuto  VinSolutions  xtime 

Regulatory Changes | Impacting EVs

Creating Uncertainty and Reshaping Competitive Landscape

Electrification

Despite years of expecting growth through innovation, government subsidies and growing consumer adoption, electrification is facing challenges.

- President signs Congressional Review Act (CRA)
- Resolutions revoking California's Clean Air Act waivers
- State-level disparities create challenges for automakers

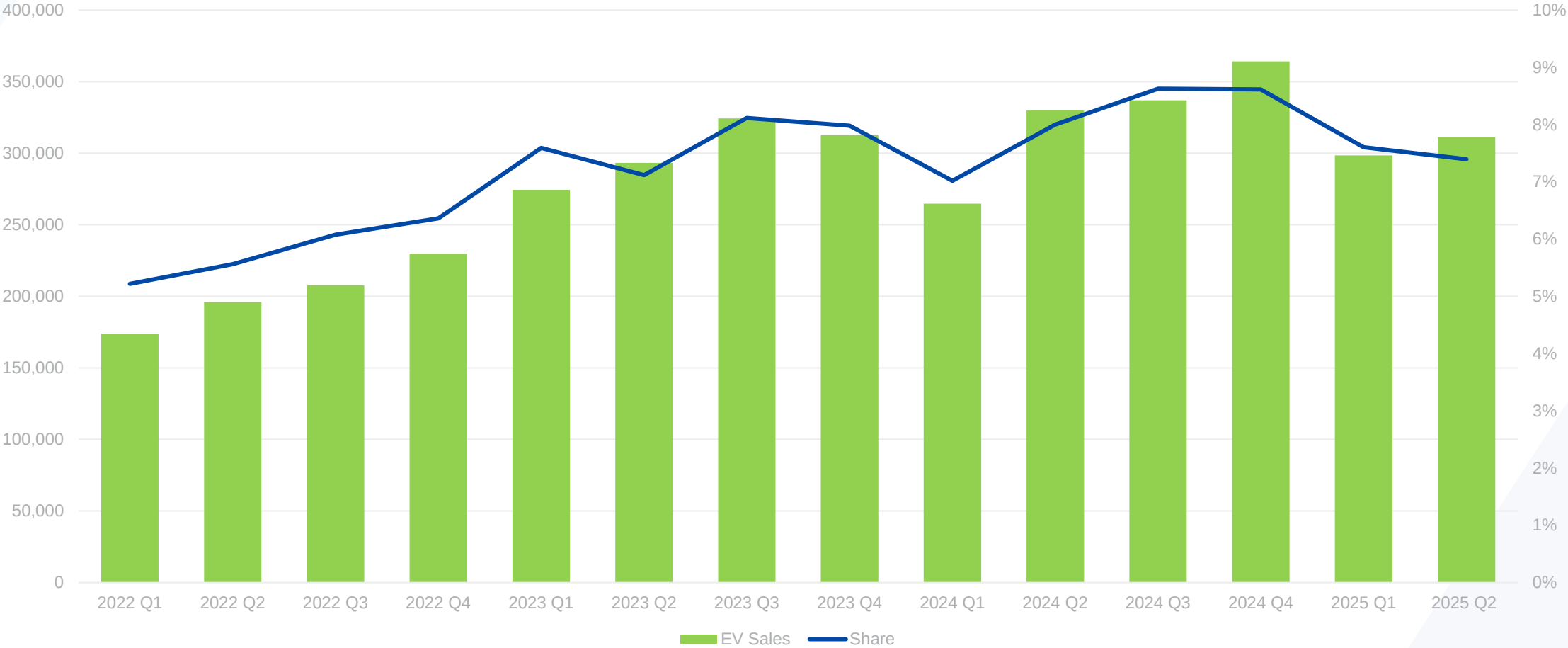
One Big Beautiful Bill

- Elimination of Federal EV Tax Credits
- Alternative Fuel Vehicle Refueling Property Credit Phased out
- Advanced Manufacturing Credit Phased Out
- **CAFE penalties** for non-compliance eliminated to Zero

Supply Chain Disruption

- Rare earth and critical mineral shortages
- Impact of Tier 2, 3 and smaller suppliers of cash flow given import tariffs imposed
- Labor constraints
- Port congestion and shipping delays

Q2 New EV Sales | Up from Q1, Lower Y-o-Y
Record Q3 Forecast as EV Credits Expire

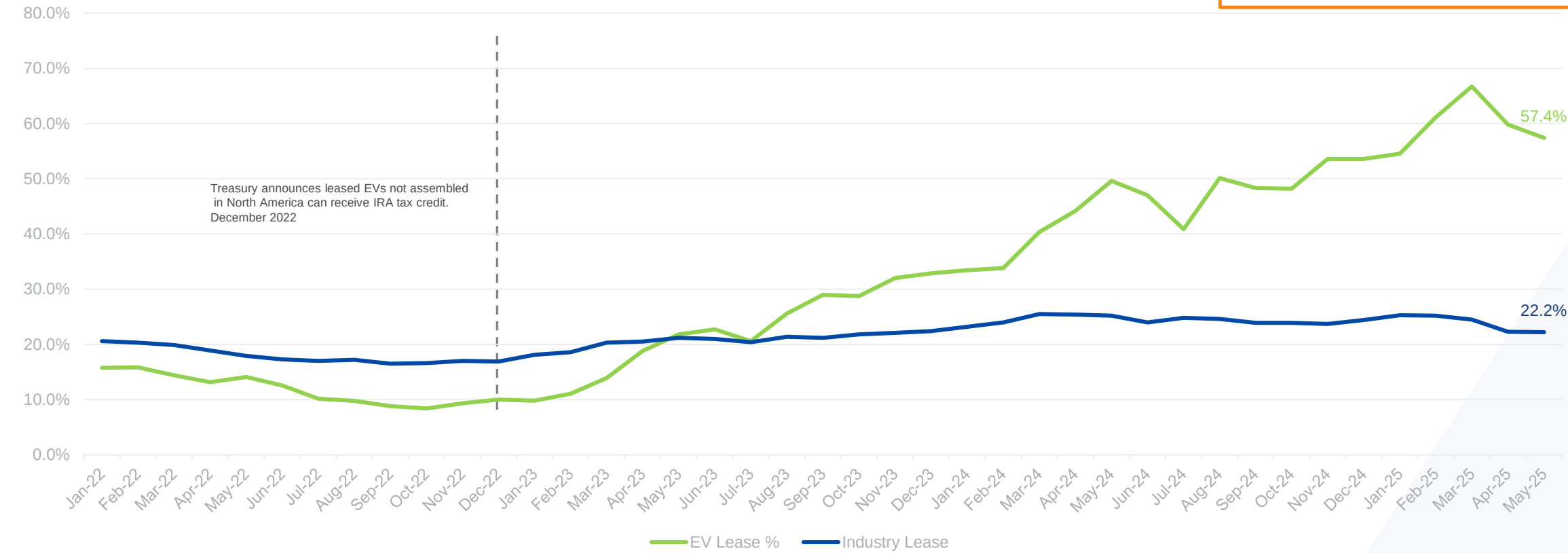


EV Leasing | Fuels EV Adoption

Nearly Triple the Industry Average

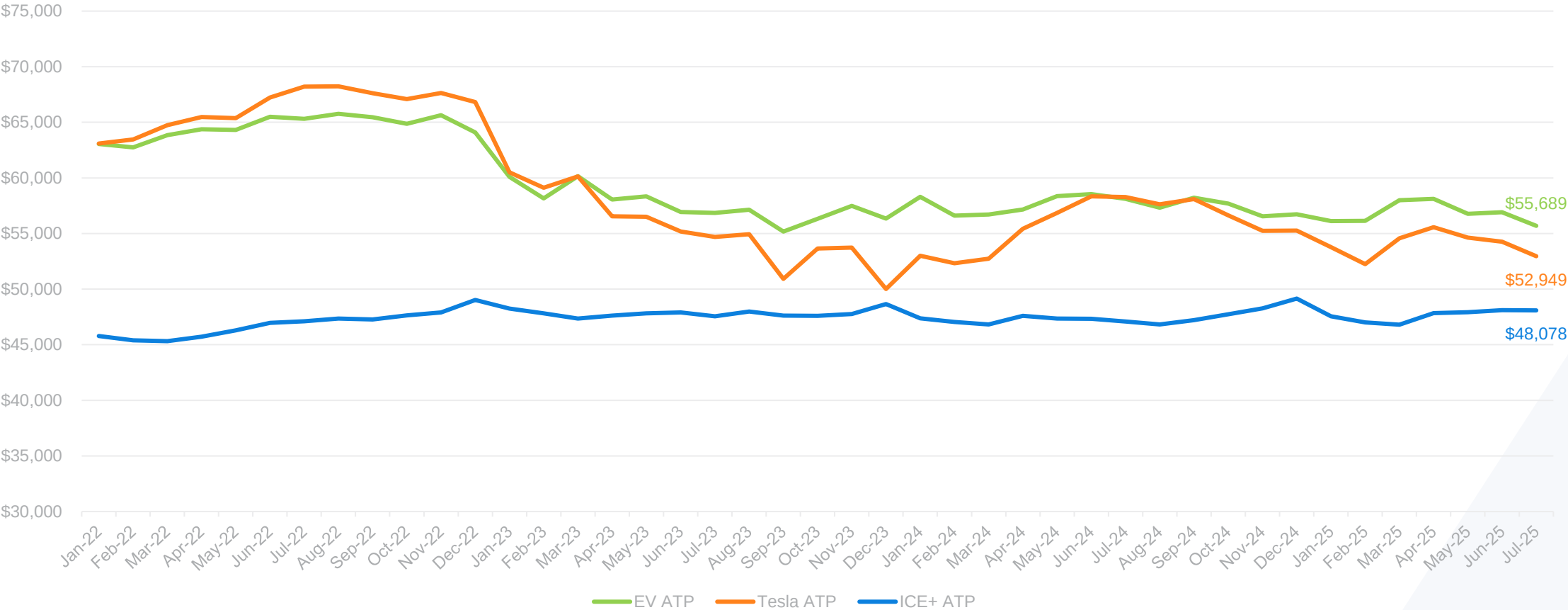
EV Lease % versus Industry

~ 1.1M EVs Leased Since 2023



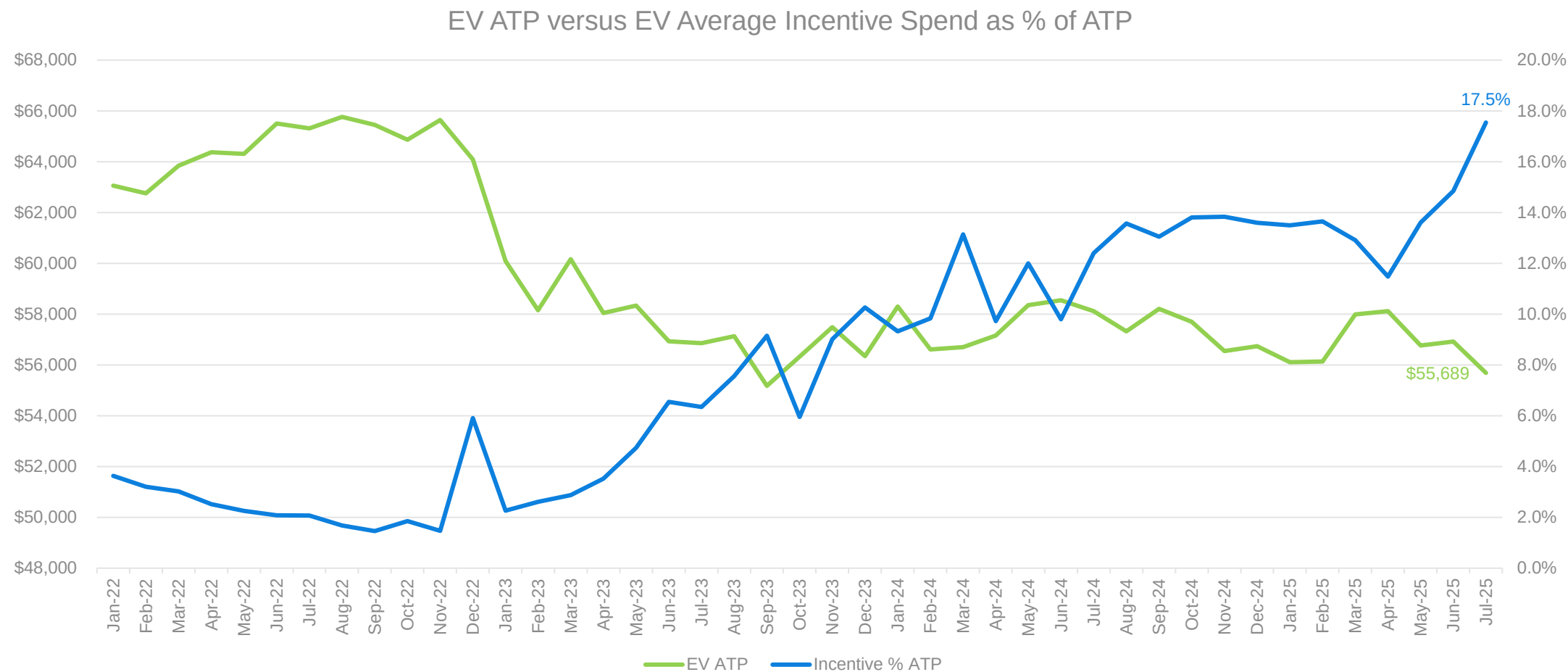
EV Average Transaction Price | Trending Down

Narrowing the price gap with ICE+ vehicles to \$7,611



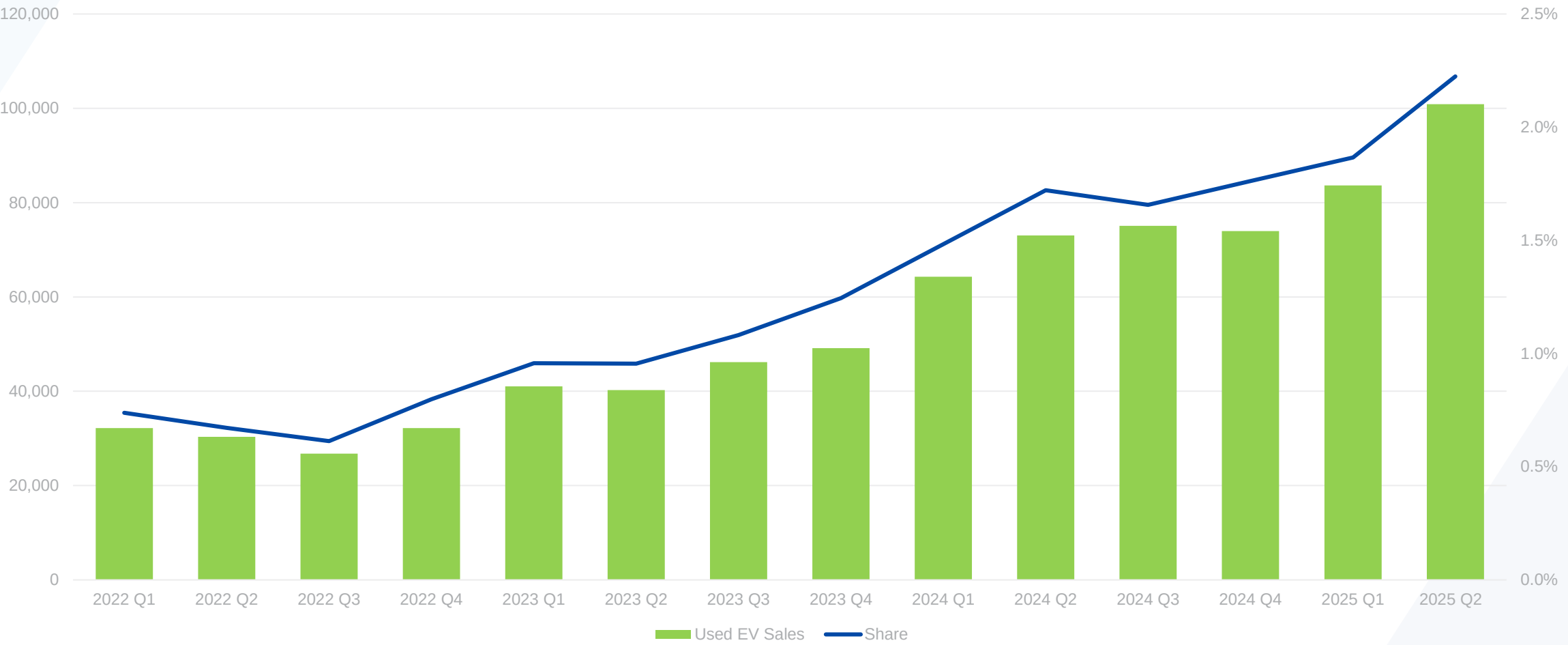
EV Incentives as % of ATP | Trending Up

In July, EV incentives reached 17.5% of ATP, totally \$9,768, while EV ATP declined – signaling aggressive discounting to drive demand and clear inventory.



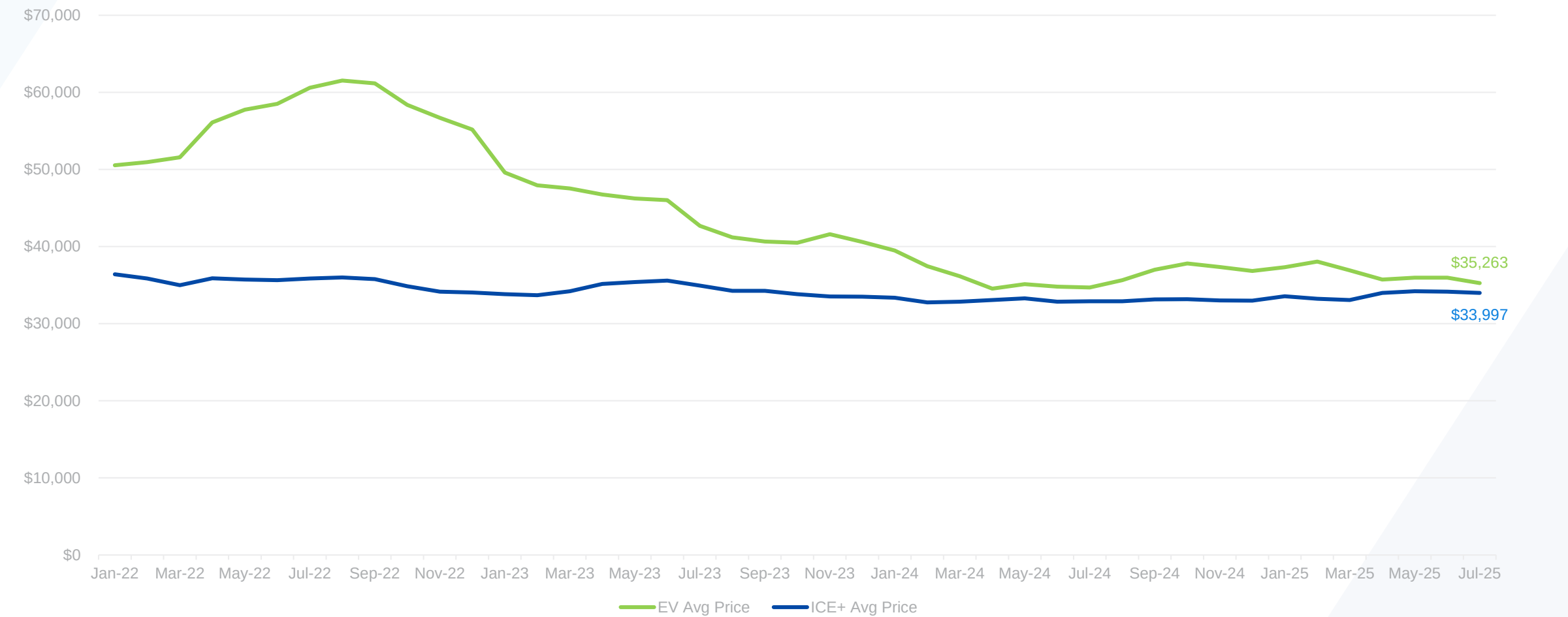
Q2 Used EV Sales | Continued Growth

Hit Record High, Exceeding 100K



Used EV Pricing | Record-Low Gap with ICE+ Vehicles

Price difference narrowed to \$1,266 in July





heliox

A Siemens Business



⚡ Vision Statement for Heliox & Siemens eMobility North America

“Be the partner of choice in North American fleet electrification by delivering best-in-class charging solutions, value, and user experience for our end customers.”





**EV CHARGER
INSTALLATIONS**

FLEET CHARGING COST REDUCTION STRATEGIES



COST: GAS VS ELECTRIC

OUR EXPERIENCE



RIVIAN R1T

- Efficiency: 2 Miles / kWh
- Electricity Cost: \$0.12 / kWh
- Monthly Mileage: 3,000
- Monthly Fuel Cost: \$180

Ford Transit 150

- Efficiency: 15 Miles / Gal
- Fuel Cost: \$2.80
- Monthly Mileage: 3,000
- Monthly Fuel Cost: \$560

MONTHLY SAVINGS: \$380 PER VEHICLE

ANNUAL SAVINGS: \$4,560 PER VEHICLE

REDUCING INFRASTRUCTURE COSTS WITH DYNAMIC LOAD MANAGEMENT

- What is Dynamic Load Management?
 - Intelligent allocation of available electrical capacity across multiple chargers
- Ideal for:
 - Office buildings and dealerships
 - Vehicles parked for 8+ hours daily
- Benefits:
 - Significantly reduces upfront electrical infrastructure costs
 - Avoids costly electrical upgrades
 - Maximizes existing power capacity
 - Scalable as fleet expands



HOME CHARGING SOLUTIONS WITH MONITORING & REIMBURSEMENT

- Alternative to Commercial Infrastructure
 - Reduces or eliminates costly commercial charging installations
- How It Works:
 - Install monitored home chargers for employees
 - Track exact kWh usage remotely
- Company Benefits:
 - Precise, automated reimbursement
 - Lower upfront capital expenditure
 - Accurate tracking of employee charging usage
- Employee Benefits:
 - Convenient, overnight charging at home
 - Easy and fair reimbursement process



THANK YOU!





ENVIRO  SPARK
LIVE. WORK. PLAY. CHARGE

● ABOUT US

Founded in 2014, EnviroSpark is one of the nation's largest charging providers and charge point operators, responsible for over 10,000 plugs deployed across the US and Canada, including over 2,500 on its own network.

EnviroSpark leverages its 11 years of experience and expertise to

- Level 2 and DC Fast Charging
- Multiple funding solutions, including:
 - Fullyfunded Network
 - Charging as a Service
 - CustomOwned

EnviroSpark's full-service approach removes complexity and lowers barriers, enabling faster deployments and long-term reliability.





WE PROVIDE A

**seamless, inhouse
solution**

FROM START
TO FINISH

Who we serve:

- Multi-family
- Retail/Office
- Hospitality
- Universities
- Municipal/State/Federal
- Mixed Use Developments
- Gas Stations/C-Stores
- Other Charge Point Operators





From the initial fleet study to smart charging management, we support your electric transition with complete turnkey solutions or customized à la carte services.



Engineering services

Feasibility studies, project management, electric fleet sizing, planning, costs and benefits, implementation strategy.



Infrastructure

Everything for EV charging: charging stations, electrical distribution equipment, terminals, etc.



Software

Smart charging management platform developed by our in-house team in Canada



Financing

Tailor-made financing programs (loans, leasing) and energy & charging monetization



by Polara



Charging reliability

Rest assured that your vehicles will always be ready and charged in time for their daily trips.



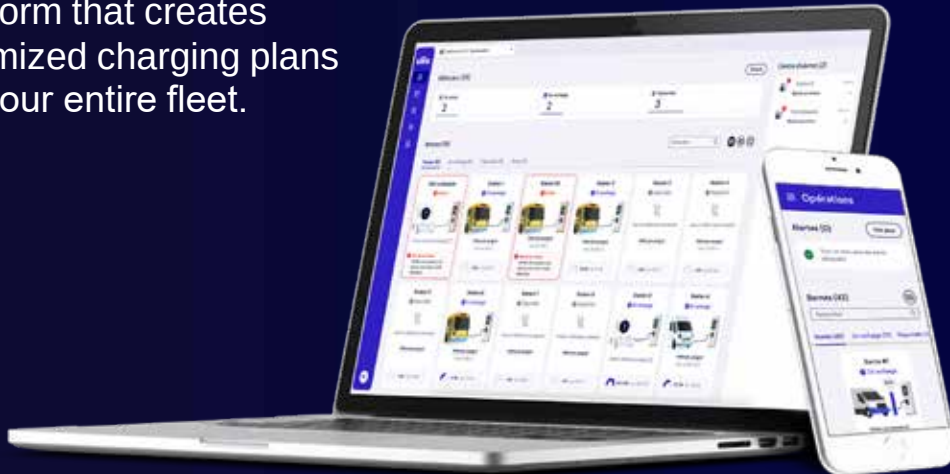
Operational assistance

Keep a constant eye on your charging operations from a centralized control screen.



Energy optimization

Save money with a platform that creates optimized charging plans for your entire fleet.



Features	Classic CMS	Cleo's platform
Remote control of charging stations (OCPP 1.6 minimum)	✓	✓
Sequential charging	✓	✓
Dynamic charging	-	✓
Agnostic and interoperable	-	✓
Centralized multi-site management	-	✓
Reduced power peaks	-	✓
Automatic vehicle recognition	-	✓
Integrations (telematics, fleet management, maintenance tools, etc.)	-	✓
Creation of charging plans / vehicle	-	✓
Automated AI route learning	-	✓
Alerts and notifications	-	✓
Reports and analysis	-	✓

We have been doing this longer than we think!

1999

Highest mpg of any automobile in its 7-year run.

We have had propulsion batteries in vehicles in the US over 25 years!

And we keep getting better at it! In the last 10 years:

- Median range tripled
- Battery cost decreased 90%
- Battery life longer than predicted



Lonnie Russell

Business Development Manager

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Polara





powerup

Georgia
Clean
Energy
Roadshow

PowerUp Mission Statement

PowerUp is dedicated to providing reliable and seamless EV charging, empowering our customers to embark on every journey with confidence. Through the deployment of innovative technology and partnerships with state and local communities, we're transforming EV travel into an effortless experience, ensuring that every charge fuels your next adventure.



PowerUp Georgia

Level 2

Max Charge Speed per Vehicle
10.0 - 19.2kW

Charging Heads
J1772

PowerUp Plugback to help communities and businesses replace broken or aging chargers



Level 3

Max Charge Speed per Vehicle
Up to 400kW

Charging Heads
CCS and NACS on each port

Partnering with communities and businesses to deploy strategic level 3 sites across the state at no cost to the host





GEORGIA NETWORK *for*
ELECTRIC MOBILITY

UNIVERSITY OF GEORGIA

ELECTRIC VEHICLES AND FIRE SAFETY

***KEY CONSIDERATIONS FOR
INDUSTRY, FLEET MANAGERS,
AND POLICYMAKERS***





GEORGIA NETWORK *for* ELECTRIC MOBILITY



GABRIELLE PIERRE
*Strategy Director, Georgia
Network for Electric Mobility*

GNEM OVERVIEW

GNEM aims to position Georgia as a regional hub for future mobility innovation through the following activities:

ACTIVITIES	MULTIDISCIPLINARY RESEARCH	COMMUNITY ENGAGEMENT	WORKFORCE DEVELOPMENT
GOALS	ADVANCE RESEARCH AND DEVELOPMENT IN ELECTRIC MOBILITY INNOVATION	ESTABLISH COMMUNITY PARTNERSHIPS FOR ADOPTION OF ELECTRIC MOBILITY ACROSS THE STATE OF GEORGIA	PREPARE THE NEXT GENERATION OF TALENT FOR THE ELECTRIC MOBILITY INDUSTRY
ON - GOING SUB - ACTIVITIES AT UGA	• Academic research partnerships • Industry R&D partnerships (incl. Manufacturing Living Labs) • Whitepapers • Market intelligence + tools	• Future Mobility Living labs • Government partnerships • EV readiness programs • K-12 programs	• Capstones • Scholarships • Training and Certificate programs • K-12 programs





 **GEORGIA NETWORK for
ELECTRIC MOBILITY**
UNIVERSITY OF GEORGIA

2025 GEORGIA EV FIRE SAFETY BRIEF

BRIDGING LOCAL POLICY IN
ATLANTA WITH A CALL FOR A
UNIVERSAL GLOBAL STANDARD



MAYOR'S OFFICE OF
Sustainability
and Resilience



EV FIRE SAFETY
2025 TECHNICAL
GUIDE



2025 GEORGIA EV
FIRE SAFETY BRIEF

EV FIRE SAFETY

2025 TECHNICAL GUIDE

 **GEORGIA NETWORK for
ELECTRIC MOBILITY**
UNIVERSITY OF GEORGIA

Did you
know?

EVs are
safer
than...

**Gasoline vehicles are
20 times more likely to catch
fire than electric vehicles.**



**You're more likely to get
struck by lightning —twice— or
have a kitchen fire than be
injured in an EV fire.**



COMPARATIVE SAFETY CHART



FEATURE	DIESEL/GAS	ELECTRIC
Fire incident rate	1 in ~1,200	1 in ~30,000–100,000
Fire cause	Fuel leaks, heat	Battery thermal runaway
Pump/Charger fires	~5,000/year (U.S.)	Very rare (<100/year est.)
Severity if fire occurs	Moderate	High intensity, rare

What actually causes an EV fire?

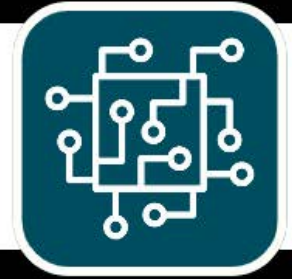
A common misconception is that EV batteries can spontaneously combust. However, data shows that battery ignition generally requires an external trigger such as:



Severe physical
impact (e.g., traffic
collision)



Exposure to sustained
high temperatures or
open flame



Significant
manufacturing defect or
electrical short circuit

These triggers can cause **thermal runaway** which occurs when a battery cell overheats uncontrollably, triggering a chain reaction in adjacent cells. This can cause fires or explosions if not contained

**EVs are significantly less likely catch
fire than gas vehicles but...
EV fires require specialized response
training and tools**

**MONITORING
AND
MAINTENANCE**

**CRITICAL
RESPONSE
TIME**

**LIMITING GAS
EFFECTS AND
CROSS
PROPAGATION**

**DECOMMISSIONING
/DISCHARGING THE
BATTERY**

**FIRE
SUPPRESSION
AND
EXTINGUISHING**

BEST PRACTICES FOR FLEET OPERATORS

Design and retrofiting



- Improve airflow/ventilation
- Separate/buffer charging areas in enclosed spaces
- Add thermal/smoke detection near chargers

Charger placement in parking decks and lots



- Avoid tight alcoves or hard to reach areas in the center of the decks where possible; these can impede first responders
- Favor near entry/exits; EVSE in dead-end corners heightens risk
- Provide signage & shutoff switches
- Limited ventilation can trap toxic gases (like hydrogen fluoride), especially in older structures lacking fire monitoring upgrades

Fire Safety training



- Facility staff & local fire crews need EV-specific response training and equipment where possible

**Many local electrical codes were
written before EVs became
mainstream**

**Updating these policies now
ensures your community or fleet is
ready for a rapidly electrifying
future**



For more info: Email gpierre@uga.edu

Workforce Development in Georgia



SAVANNAH
TECHNICAL COLLEGE

Fireside Chat

Cynthia Cradduck – STC- Moderator

Tal Loos – STC

Karisa Young - HGMA



Workforce Development for the E-Mobility Sector

Meeting the Demand



- § CGTC just completed a \$6.5M renovation at the Milledgeville campus
- § This included an overhaul of our Automotive Technology Program to begin delivery of EV & hybrid vehicle content



Exciting New Programs



- § CGTC will begin offering two EV & hybrid related Technical Certificates in Fall 2025
- § Electric Vehicle Professional (EVP1 Technical Certificate)
- § Hybrid / Electric Vehicle Repair Technician (HVR1 Technical Certificate)



Growing Partnerships



- § The Electric Vehicle Professional Program will also be a Dual Enrollment option at Hutchings Innovation & Technology Academy
- § CGTC continues to partner with Rivian to host EV related training for technicians, first responders, etc.



Questions / Comments

E-Mobility and the Workforce Pipeline

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Regional Transportation Electrification Plan

Clean Energy Roadshow

Folashade Alao, PhD

Principal, Transportation Planner
August 14, 2025



ONE
great
REGION



The Atlanta Regional Commission is the federally required Metropolitan Planning Organization (MPO) for a 19-county region which is currently home to about 6.1 million people. This is expected to grow to 7.9 million by 2050.

Regional Transportation Electrification Plan

- The purpose of this plan was to understand the regional EV ecosystem and develop a policy strategy and approach.
- Primary focus on Light Duty (LD) Vehicles
- Understand vehicle ownership in the region and forecast potential growth
- Understand current EV infrastructure and forecast future charging infrastructure needs
- Examine barriers to EV adoption in the region
- Identify actions to strengthen EV ecosystem in the region, e.g. workforce development and economic development

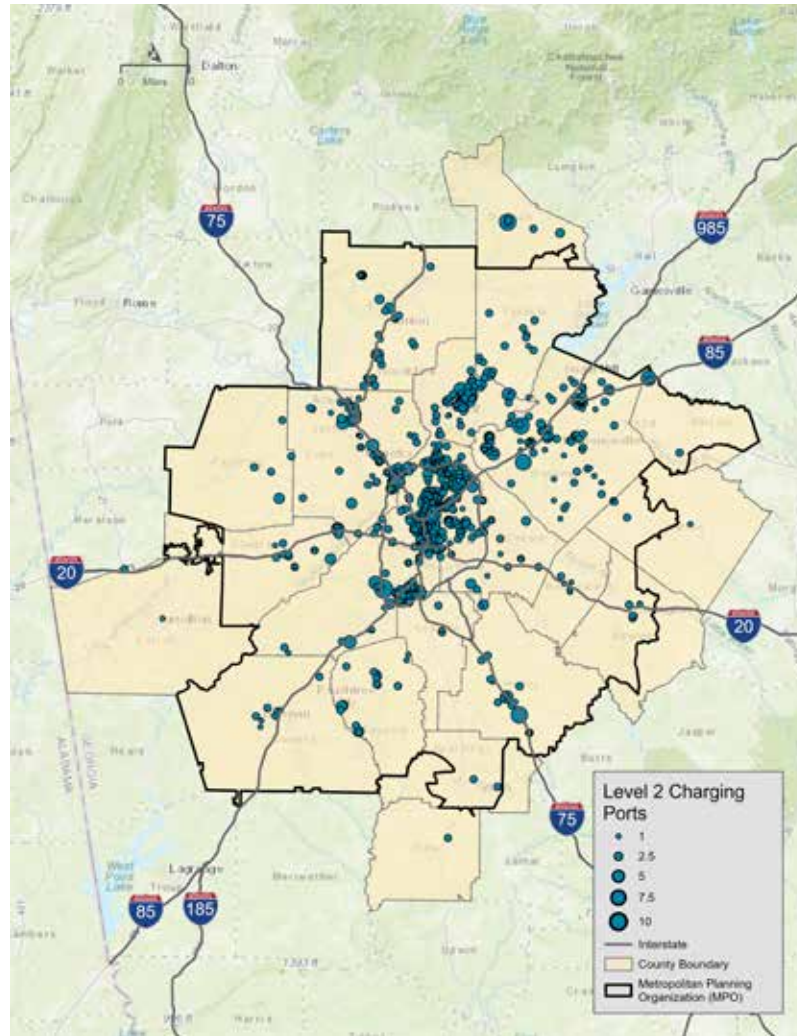
Key Vehicle Data for Metro Atlanta GA

Total # of Registered Vehicles in Metro Atlanta (8/25)	5,415,684
Total # of Registered EVs in Metro Atlanta (8/25)	110,498
Total projected # of EVs in Metro Atlanta fleet by 2030 (Based on 50% annual sales rate in 2030)	515,000
Level 2 (shared private and publicly accessible)	Additional Ports Needed: 47,100
DCFC	1,800

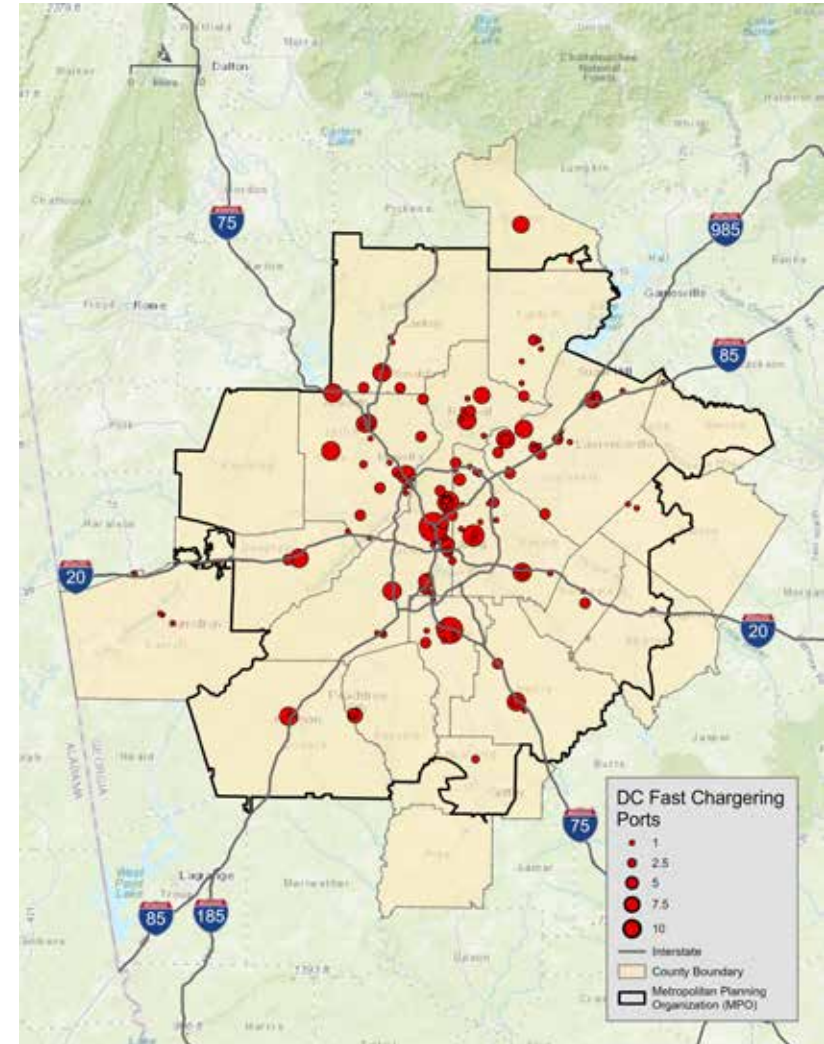
Existing Charging Infrastructure

- Existing Level 2 Chargers and Level 3 DCFC (fast chargers) are clustered centrally within Atlanta and suburban areas like North Fulton, northern DeKalb, Cobb and Gwinnett.
- Current gaps exist south of I-20.

Level 2 Charging Infrastructure



Level 3 DCFC Charging Infrastructure



Implications for Local Government

Public Fleet Transition- Fleet conversion to support reduced fuel and maintenance costs (Cobb County, DeKalb County, and City of Brookhaven as well as MARTA, and ATL Xpress).

Public Charging- Boost economic development, e.g. retail, restaurants, and downtown areas as well as address range anxiety (City of Peachtree Corners). Address equity considerations, particularly for multi-family housing.

Codes- Review codes like zoning, building, permitting to support safe and seamless charging infrastructure (City of Atlanta).

Partners and Coordination- Work with new partners including utilities, commercial property owners, business owners, and charging service providers (Georgia Power, EMCs).

Technical Knowledge- Increase understanding of how transportation electrification crosses multiple government departments.

Funding- Explore opportunities to facilitate accelerated deployment.

Regional Medium/Heavy-Duty Alternative Vehicle Alternative Fuels Plan

- ▮ **Background:** Recommendation from ARC's *Regional Transportation Electrification Plan* and *Atlanta Regional Freight Mobility Plan*.
- ▮ **Goals:** Improve fuel efficiency and air quality, utilize national resources, foster energy independence, and stimulate the local economy
- ▮ **Description:** This plan will examine alternative fuels infrastructure (including electrification, hydrogen, natural gas, biodiesel, and propane) for medium and heavy-duty vehicles within the metro region. The plan seeks to:
 - Analyze alternative fuels infrastructure needs and identify gaps
 - Provide recommendations and an implementation guide for expanding alternative fuels infrastructure
 - Identify pilot project locations for potential alternative fuels infrastructure



Hyundai XCIENT fuel cell trucks at the Port of Brunswick, Georgia

Source: Hyundai



Thank You

Folashade Alao, PhD

Principal, Transportation Planner

falao@atlantaregional.org



SCAN ME



EV • STRATEGIES • IN • MIDDLE • GEORGIA

CLEAN ENERGY ROADSHOW – MACON, GA

AUGUST 13, 2025



“ WE WILL ACT NOW TO REDUCE NO_x AND OZONE IN THE MIDDLE GEORGIA COMMUNITY TO ENSURE THAT WE PROTECT THE REGION'S PUBLIC HEALTH, MAINTAIN A ROBUST ECONOMIC GROWTH, AND SHOW LEADERSHIP IN ENVIRONMENTAL SUSTAINABILITY. ”

Middle Georgia Clean Air Coalition Charter

2025 Priorities

Goal #1: Prepare for Electric Vehicle & Alternative Fuel Transitions

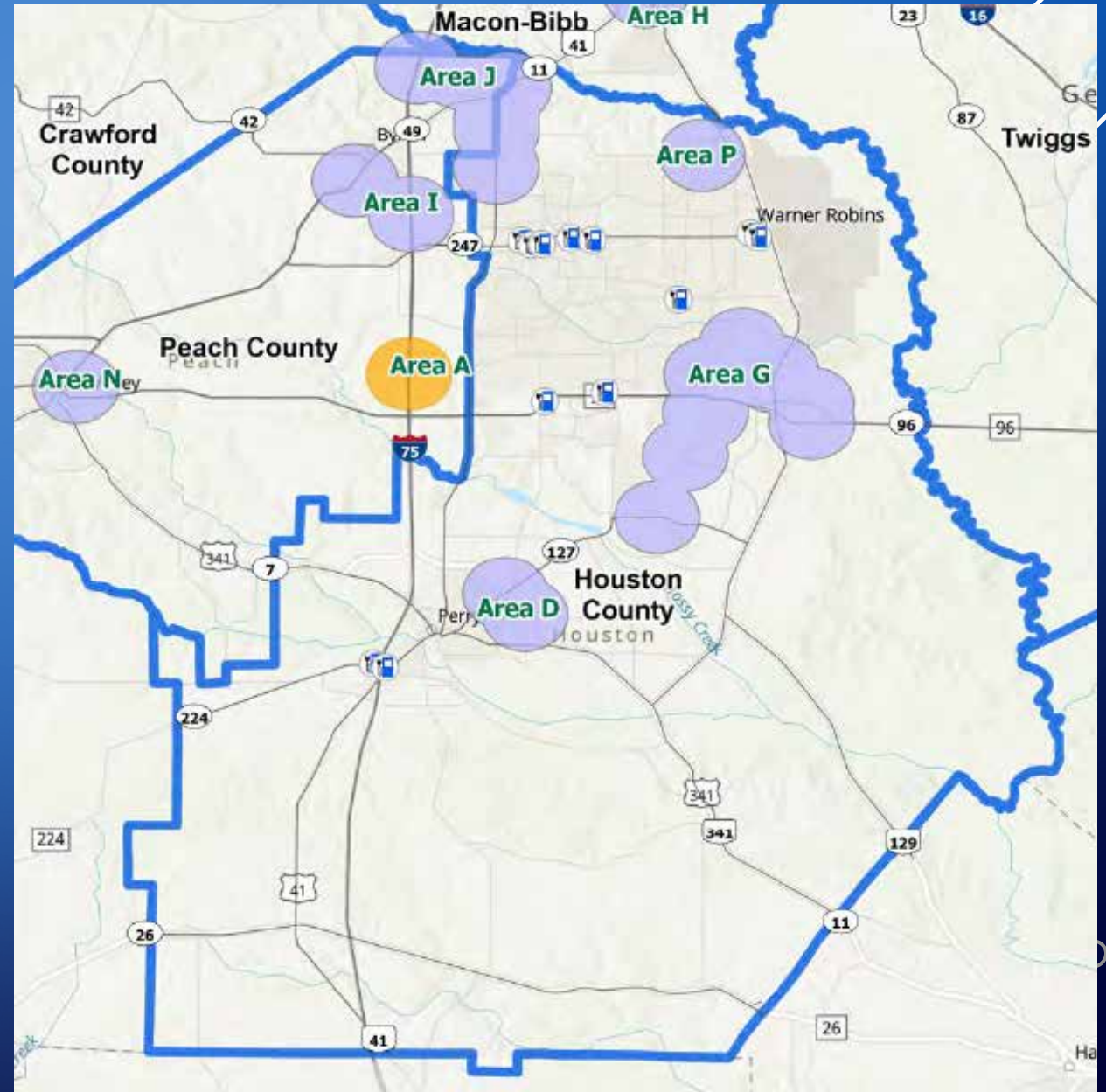
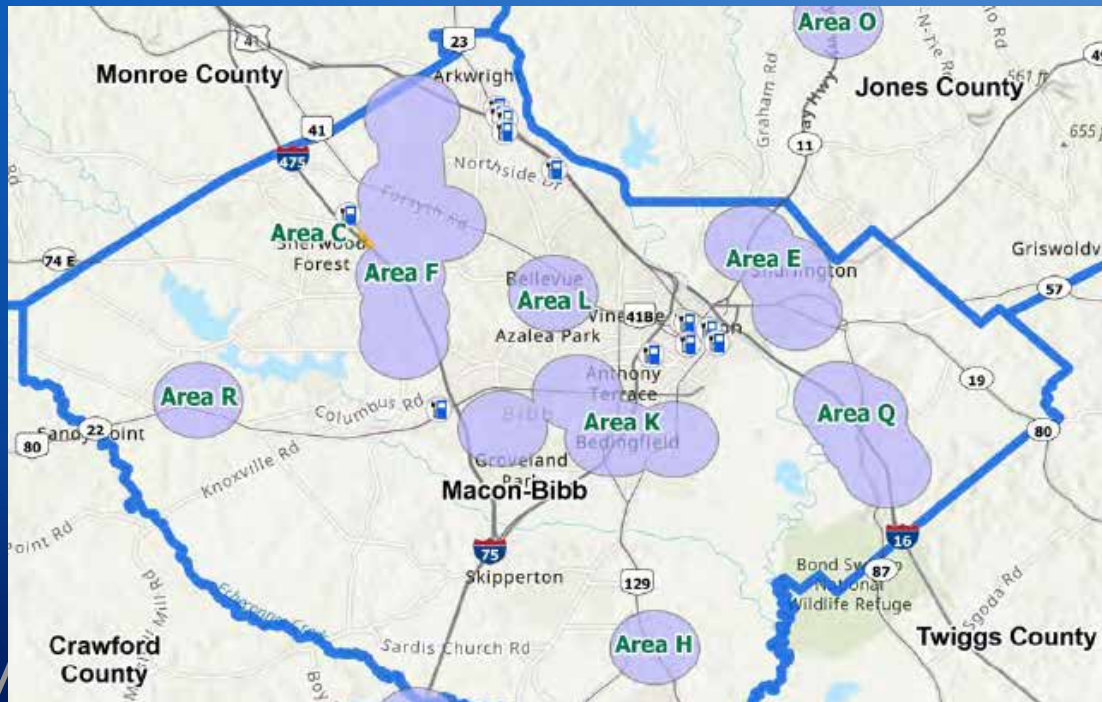
Goal #2: Expand Middle Georgia Solar Programs

Goal #3: Undertake Projects for Green, Efficient, and Healthy Buildings

Goal #4: Support Transit and Smart Growth Planning to Improve Air Quality

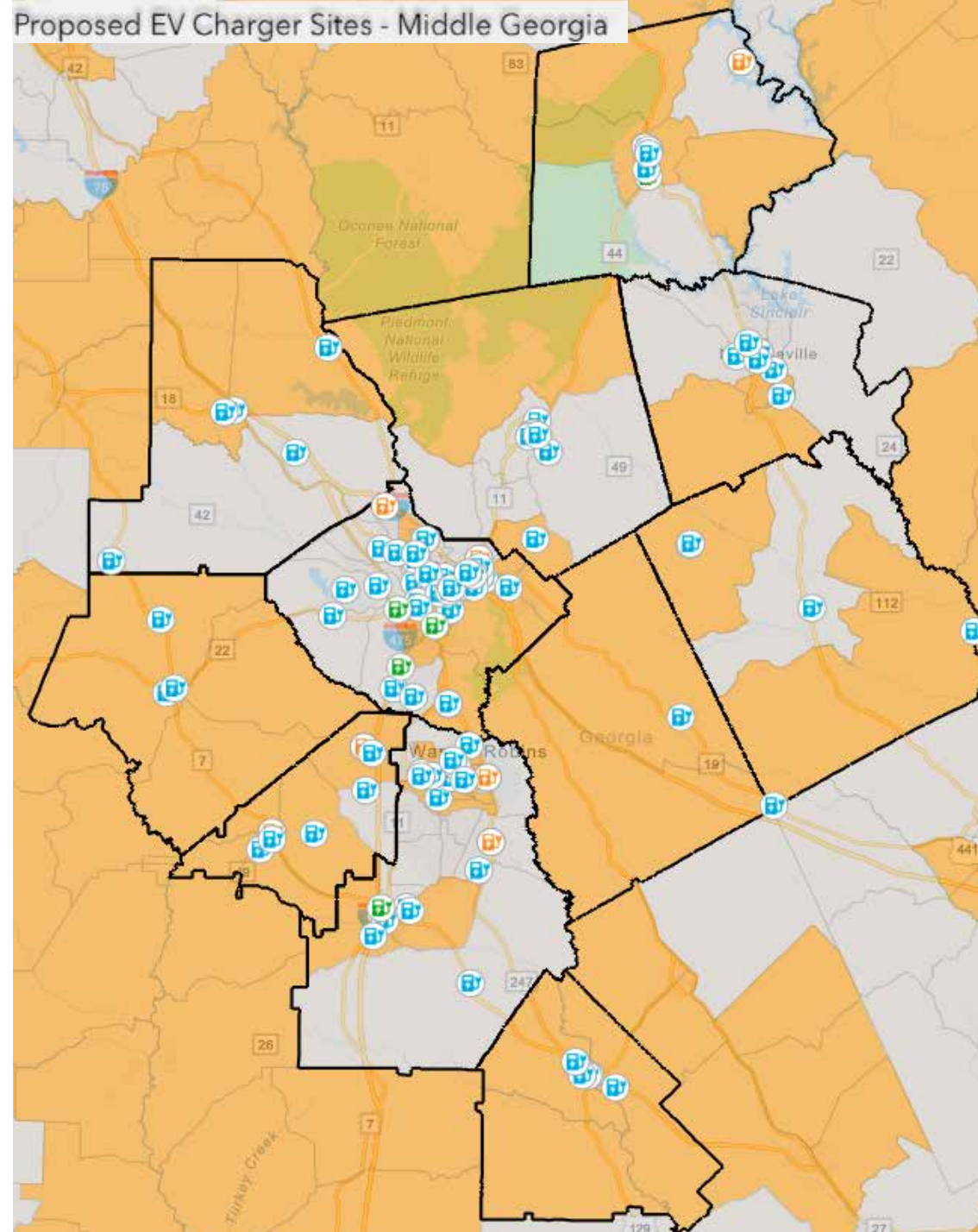


REGIONAL EV PLANNING



CFI Program Application

- Approx. 100 Proposed Charger Sites
 - Every Middle Georgian within 15 minutes of an EV Charger
 - 54% Transportation Disadvantaged
- Awarded \$15M in August 2024 (Round 1B)
- Pending Grant Agreement w/ USDOT





FOR•MORE• INFORMATION:

- Visit our website:
www.middlegacleanair.org
- Greg Boike, MGCAC Project Manager:
 - gboike@mg-rc.org | 478-722-6945



Clean Cities and
Communities



Alternative Transportation Funding and Resources

Designated Clean Cities and Communities Coalition, a U.S. Department of Energy partnership to foster transportation energy choices nationwide

How We Can Help

Advance local alternative transportation initiatives for all communities through:

- Fleet conversions: match the right fuel for the right application
- Grant partnership & management
- Education and outreach



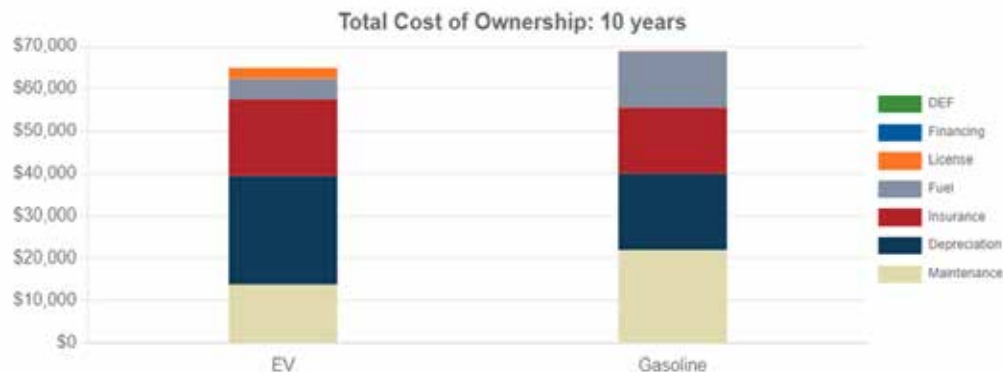
Funding Landscape

- **DOT Lo or No Emission Grant** - vehicle replacement, released annually in Spring (closed in summer 2025)
- **DOE Vehicle Technology Office FOA** - released annually for new & emerging transportation technologies
- **EPA Diesel Emission Reduction Act:** Federal and State (school buses only), potential for reduced funding in 2026
- **EPA Clean School Bus Program:** future uncertain
- **Clean Cities and Communities:** 2026 budget includes competitive grants

Current Incentives

- **Alternative Fuel Excise Tax Credit** - a tax credit in the amount of up to \$0.50 per gallon for alternative fuels (not yet renewed)
- **Federal Clean Vehicle Tax Credits** still available through September 30, 2025. Up to \$7,500 new and \$4,000 used.
- **EPD Clean Vehicle Charger and Converted Vehicle Tax Credits** (+ Direct Pay): 10% or up to \$2,500 per charger or per converted vehicle (CNG, EV, Propane, Hydrogen)
- Georgia Power Make Ready Program and Business Rebate Program
- Local EMC or MEAG Incentives

Available Tools: AFLEET



Welcome To AFLEET

The Department of Energy's Technology Integration Program has enlisted the expertise of Argonne to develop a tool to examine both the environmental and economic costs and benefits of alternative fuel and advanced vehicles (AFVs). Argonne developed the Alternative Fuel Life-Cycle Environmental and Economic Transportation (AFLEET) Tool to help stakeholders estimate petroleum use, greenhouse gas (GHG) emissions, air pollutant emissions, and cost of ownership of light-duty, heavy-duty, and off-road vehicles. AFLEET can be accessed via spreadsheet and online versions. In addition, the ATRAVEL Tool has been built using AFLEET data to examine the costs and benefits of different modes for personal travel.

AFLEET Tool (xlsx) The AFLEET spreadsheet provides detailed energy, emission, and cost data for light-duty, heavy-duty, and off-road AFVs. It has 10 calculations depending on the user's goals including: - Simple payback - Total cost of ownership - Fleet footprint - Idle reduction - Electric vehicle charging and AFV fueling infrastructure	AFLEET Online AFLEET Online replicates five of the spreadsheet's calculators: Payback, On-Road, Payback Off-Road, TCO, EV Rate, and Change TCO with a user-friendly interface and analyzes the following metrics: - Petroleum use - Greenhouse gas emissions - Air pollutant emissions - Simple payback - Total cost of ownership - Cost of EV charging	HDVEC The Heavy Duty Vehicle Emissions Calculator (HDVEC) is an AFLEET-based online tool that compares NOx, PM, CO, and funding cost-effectiveness of environmental mitigation projects for the following fuel types: - Diesel - Electric - Natural gas - Propane	ATRAVEL The ATRAVEL Tool was developed to estimate costs, travel time, and emissions of private vehicle ownership and other travel modes based on your location and travel patterns, while also providing related travel metrics at both local and regional levels. The travel modes currently included are: - Private vehicle - Transit - Ride-hail	AFLEET CFI The AFLEET Charging and Fueling Infrastructure (CFI) Emissions Tool estimates GHG and air pollutants emissions for proposals to the FHWA's CFI Discretionary Grant Program for the following fuel types: - Electric - Hydrogen - Natural gas - Propane
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How We Help Fleets

IS MY FLEET A GOOD CANDIDATE?



Goals: Your fleet is looking for ways to save on fuel, operations, and maintenance costs, as well as reduce pollutant emissions to support public health.



Fleet Age: Your fleet includes vehicles that are scheduled for replacement.



Data Driven: Your fleet is interested in employing data-driven analysis to make procurement decisions.



OUR PROCESS

Review Current Fleet: We help identify your fleet's needs and propose potential vehicle replacement solutions.

Determine Ideal Vehicle/Fuel Options: We assess which alternative fuel(s) meet the needs of each fleet vehicle and application.

Create a Conversion Plan: We help calculate your fleet's total cost of ownership (TCO) savings and create your vehicle replacement plan. This includes determining new charging/fueling needs.

Review Funding Opportunities: We research funding, incentives, and rebate opportunities to help offset vehicle replacement and fueling infrastructure costs.



Clean Cities and
Communities



Thank you!

Frank Morris, Executive Director
frank@cleancitiesgeorgia.org

Sumner Pomeroy, Program Manager
sumner@cleancitiesgeorgia.org

For more information visit:
www.cleancitiesgeorgia.org

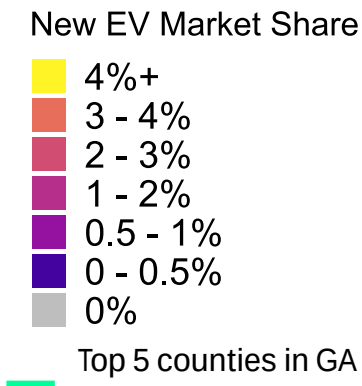
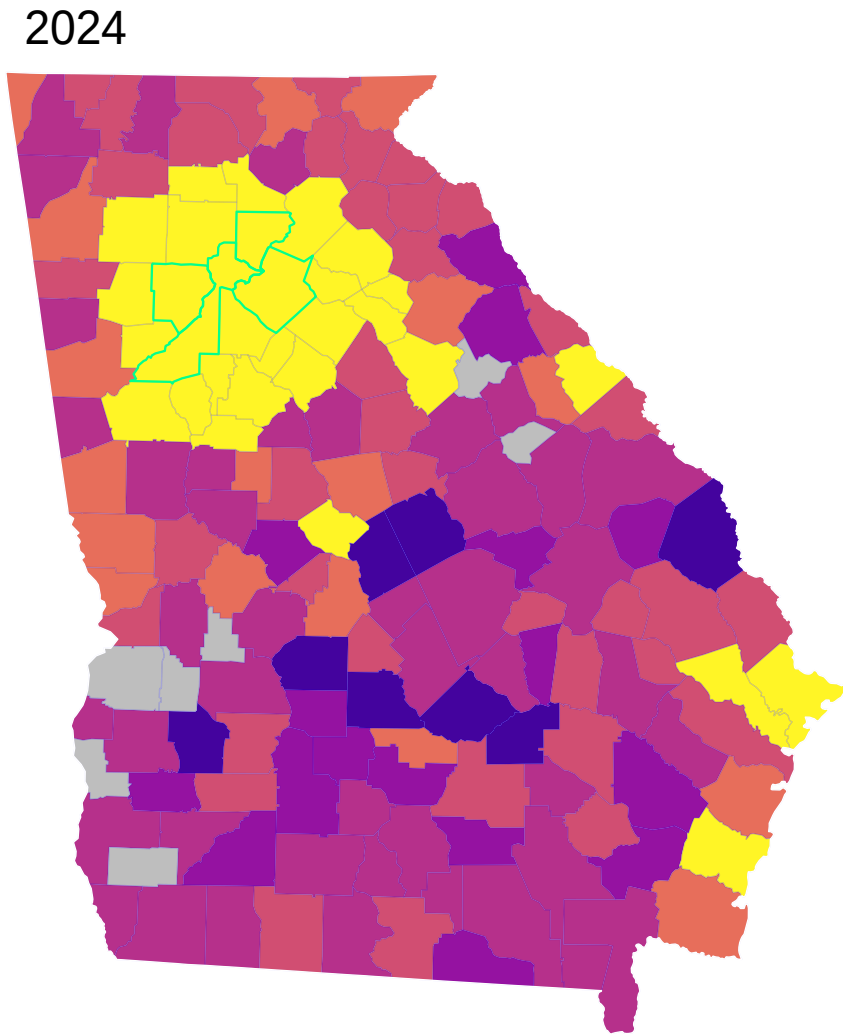
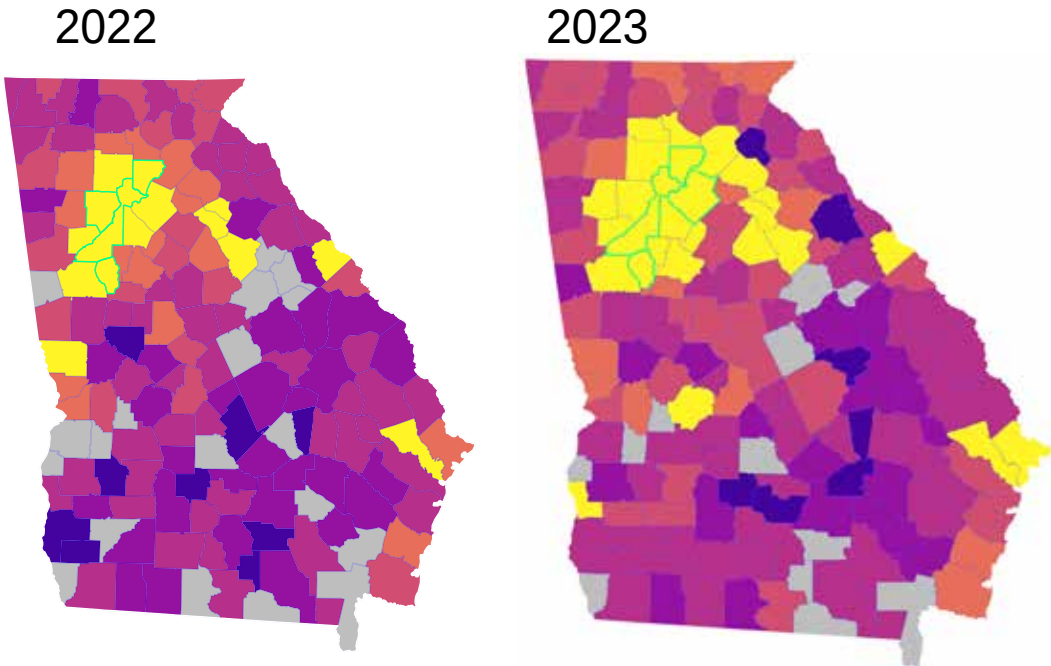
Georgia Power Electric Transportation



Georgia New EV Market Share

YE'20	New vehicle market share was 1.25% in GA
YE'21	Nationwide avg new EV market share: 4.3% State avg new EV market share: 2.9%
YE'22	Nationwide avg new EV market share: 7.3% State avg new EV market share: 5.4%
YE'23	Nationwide avg new EV market share: 9.2% State avg new EV market share: 7.4%
YE'24	Nationwide avg new EV market share: 10.1% State avg new EV market share: 7.8%

New EV Market Share:
New EV sales (or registrations) for a given
timeframe (i.e., Jan. 2024 – Dec. 2024)



Where to Start

Engage your Utility

- Capacity, System Upgrades, Supply Chain, Transformer Location
- Rates & Resiliency: Fueling Strategies, Back-up Generation, Battery Storage
- Funding Programs: Make Ready, Rebates

Charging Master Plan

- Residents, Employees, Visitors, Fleets
- Sustainability Plans
- Wants vs. Needs

Know your Community

- Municipalities
- Ordinances, permitting, etc.
- Private Companies

Chargers Types & Capabilities

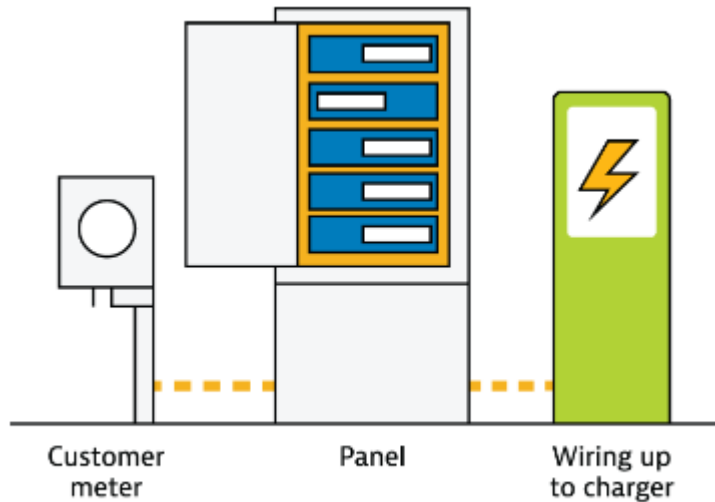
- Research: Capabilities, Uptimes, Warranties, Costs, Data Analytics
- Maintenance Planning
- Level 2 & DCFC

Funding Opportunities

- Federal, State & Local
- Utilities
- Access to a Professional Grant Writer

Programs for Business Customers

Make Ready Infrastructure Program



Georgia Power will install and maintain electrical infrastructure behind your meter up to your chargers.

- Must be a public-facing installation (accessible or serving the public)
- Must install 6 or more Level 2 chargers or at least 1 DCFC

EV Charger Plus Rebate Program



Georgia Power offers rebates to business customers installing EV chargers and associated infrastructure.

- Rebate amounts vary based on the type, kW output, and number of chargers installed.
- Receive up to \$30,000 per project. Pre-approval may be required.

Vehicle to Grid Pilot

Georgia Power will study V2G technology and its potential impacts to customers and to the System.



Installation of up to
**10 Bi-Directional
Chargers**

Identifying up to
**3-4 Public School
System Locations w/
Electric School Buses**

GOALS

Understand System
Flexibility, Resiliency, &
Economics

Document
Interconnection
Processes

Why School Buses?

School buses have
regular, predictable
schedules

School buses have large
batteries that can store
significant amounts of
energy.

School systems have already
purchased electric buses
and interest in V2G

Electrifying school buses
reduces more greenhouse
gas (GHG) emissions





Georgia
Power

georgiapower.com/et

Why Autogas?

Mark Denton

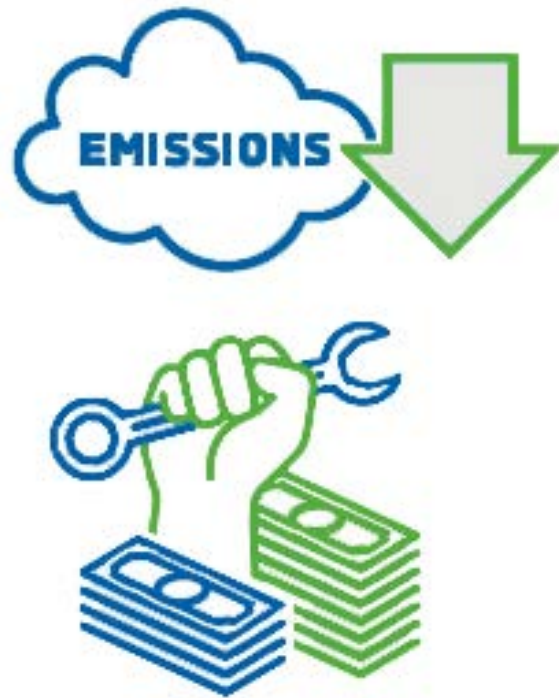
V P, Business Development

Alliance Autogas

mldenton@allianceautogas.com



What Makes an Alternative Energy Adoption Successful?



- Reduced emissions without increasing cost or losing efficiency.
- TCO reduction or ROI realized before the end of the lifecycle.
- Similar (or better) performance than the original fuel without compromising range.
- High-volume supply of energy domestically sourced.

Alliance Refueling Infrastructure



Customers can choose private, on-site refueling infrastructure scaled for their needs, or take advantage of existing public or private refueling networks.



Triton Dispenser

- 24/7 access with AAG Fuel Card
- Safe to use, no PPE required
- Transactions and fuel usage reporting, organized by vehicle
- Centralized location for all vehicles

MOBILE COUNTY PUBLIC SCHOOL SYSTEM

Fuel Cost Analysis (as of 08/01/25):

AAA cost of Diesel (minus taxes) in Mobile, Al = \$2.93 per gallon

Cost of propane to Mobile County Schools= \$1.29 per gallon

Savings \$1.64 per gallon

Assumptions based on Mobile County Data:

Diesel school bus averages 6.5 miles per gallon

Propane school bus averages 4.5 miles per gallon

Assuming 12,000 miles per year:

Diesel- $12,000 / 6.5 = 1846$ gallons x \$2.93= \$5,409.00 per year

Propane- $12,000 / 4.5 = 2666$ gallons x \$1.29= \$3,439.00 per year

ANNUAL FUEL COST SAVINGS= \$1,970.00 PER YEAR

ANNUAL MAINTENANCE COST SAVINGS= \$1,020.00 PER YEAR

ANNUAL TOTAL COST SAVINGS PER BUS= \$ 2,990.00 PER YEAR

ADKISON TOWING

TYPE OF FLEET:

TOWING & ROADSIDE

RECOVERY SERVICES

AAG FUELING PARTNER

BLOSSMAN GAS - JACKSONVILLE, FL

NUMBER OF VEHICLES

10 CONVERTED OUT OF 26

TOTAL ESTIMATED ANNUAL SAVINGS

\$191,200.00

ANNUAL MILES PER VEHICLE

80,000

AUTOGAS FUELING STATION

ON-SITE

TIME OPERATING ON AUTOGAS

2 YEARS



Mark Denton

Vice President, Business Development

Cell: 251-610-2307

email: mldenton@allianceautogas.com

Blue Bird Vision Propane (Type C)

ROUSH[®]
CLEANTECH

Engine

7.3L V8 Ford Engine with ROUSH CleanTech propane fuel system

Applications

169" / 189" / 217" / 238" / 252" / 273" / 280" wheelbase configurations

6-speed automatic transmission

Fuel Tank Capacity

- **Short:** 47 gallons (usable)
- **Standard:** 67 gallons (usable)
- **Extended:** 93 gallons (usable)

Technical Specifications

- EPA and CARB approved
- GVWR: 33,000 lbs.
- Up to 77 passengers



BLUE BIRD[®]



13+ Year Partnership



§ Exclusive partnership

§ OEM propane build



BLUE BIRD®

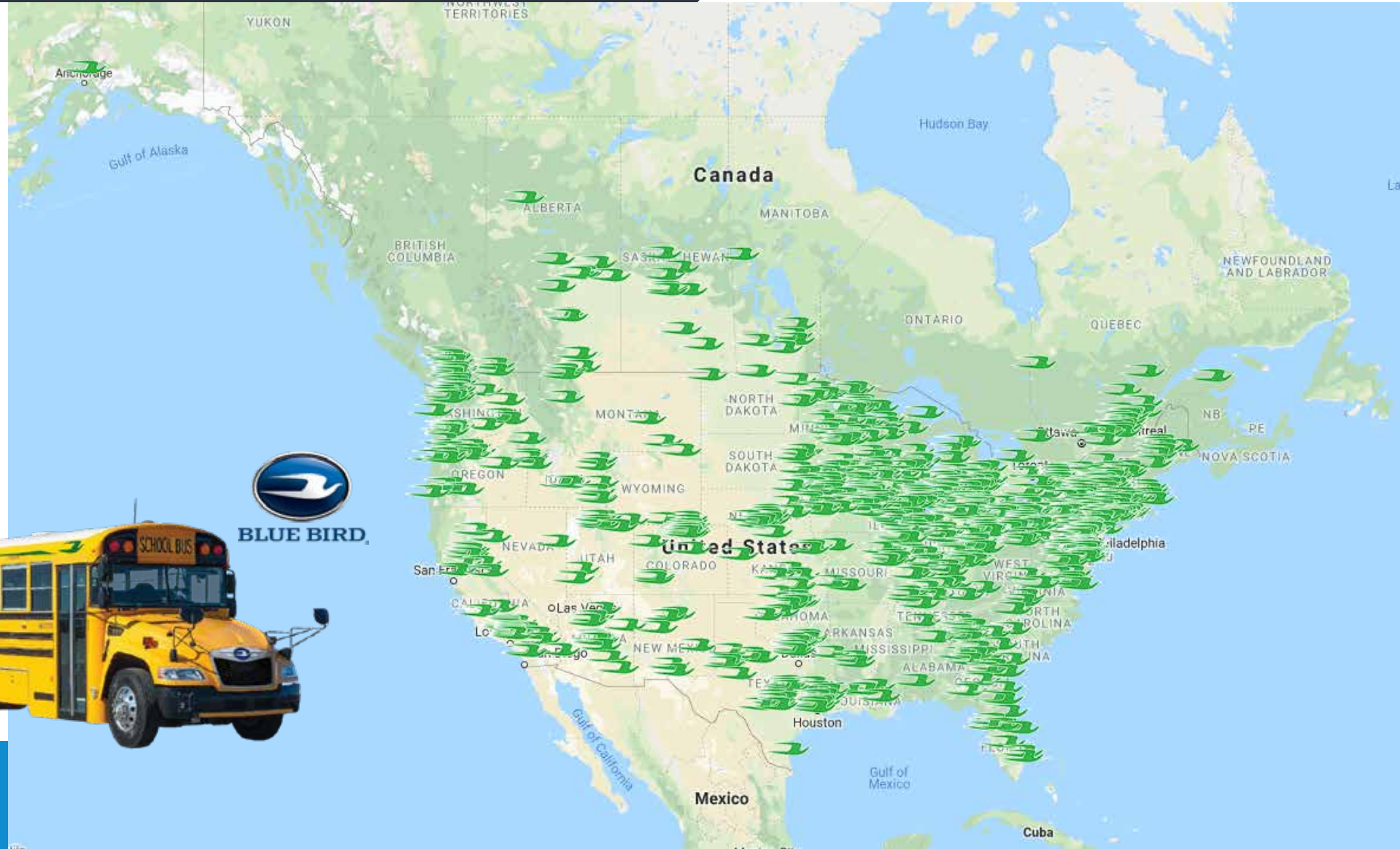


ROUSHcleantech.com | 800.59.ROUSH

Propane School Bus Deployments

ROUSH[®]
CLEANTECH

More than
22,500
on the road



Top 100 School Bus Fleets Adopt

ROUSH
CLEANTECH



36%

of the largest 100 school district fleets in the country operate a **Blue Bird** propane school bus



ROUSHcleantech.com | 800.59.ROUSH

Propane Buses Driven to Customer Locations



Zero & Near Zero Powertrain Energy Options



- Nearly 2,500 in school districts in 41 states
- Up to 150 mile range
- Less maintenance costs than diesel
- Infrastructure equipment varies:
 - level 2 - \$3-5k
 - DC Fast \$15-25k



- Over 22,500 in over 1100 school districts nationwide
- Over 400 mile range
- Average fuel/maintenance cost savings over diesel = \$3,700/bus
- Infrastructure equipment provided by fuel supplier

7.3L V8 Engine Stats

LEANER

CLEANER

MEANER



7.3

Engine RPM



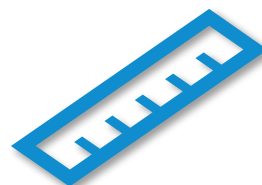
Idle: 680 / Max: 4,050

Power



HP: 335 / Torque:
468 ft-lbs. @ 3,900 RPM

Design



90° V8 / 445 CI / Pushrod 2V

Compression



10.5 to 1

Fuel & Maintenance Cost Comparison of Propane vs Diesel

Bibb County, GA							
Propane Buses					Fuel Cost	Maint. Cost	Total
12 month total miles with propane buses				2,053,143			
12 month total cost					\$790,599	\$456,716	
Cost per mile on propane					\$0.39	\$0.22	
Diesel Buses							
12 month total miles with diesel buses				393,830			
12 month total cost					\$259,640	\$179,148	
Cost per mile on diesel					\$0.66	\$0.45	
Savings/mile operating on propane					\$0.27	\$0.23	\$0.51



YANCEY
Bus Sales & Service

Ø Propane Autogas
Ø Electric



*Georgia's Green School
Bus Partner*

Considering adding Alternative Fuel School Buses into your fleet?

- Ø Longevity – Roush Partnership
- Ø Sustainability – Long Term Purchases
- Ø Total Cost of Ownership (TCO) – Repair & Maintenance/Fuel Price
- Ø Infrastructure
- Ø Funding Opportunities – EPA Rebates & Grants
 - Ø YBSS can guide districts through the process.



Trey Studstill
Director | Alternative Power
trey_studstill@yanceybus.com



Sam Ham

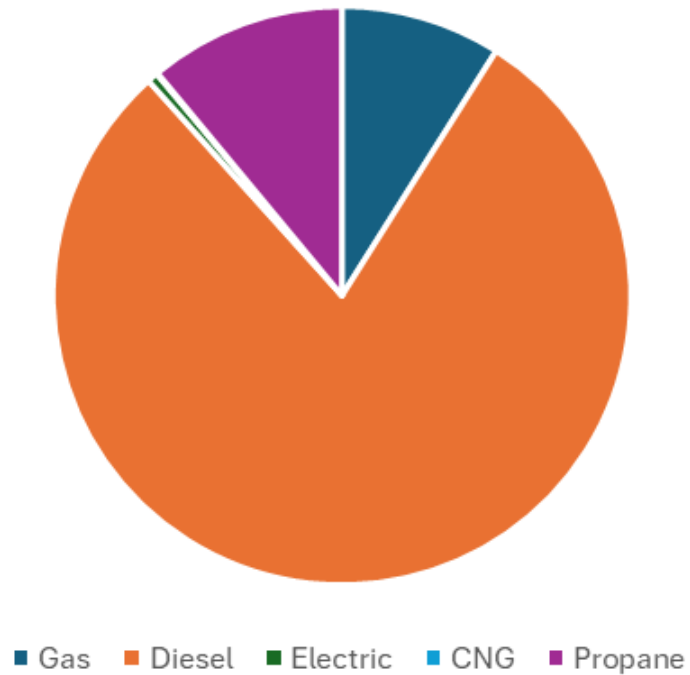
Executive Director

Why is GAPT here?

MISSION: “GAPT will build capacity in pupil transportation professionals by providing information, promoting innovation and serving as an advocate for those professionals as they partner with their communities to meet the needs of their students.”

- ü Professional Development for Transportation Directors and Technicians
- ü Provide Information on School Bus Fleet (a.k.a. Yellow Fleet)
and Support Vehicles (a.k.a. White Fleet)
- ü School Bus Fleet Federal Funding Revisions... Now What?

Fuel Type 1989-2025



Georgia School Buses Model Year 1989 – 2025

20,241

- **Gas** **1,813**
- **Diesel** **16,064**
- **Electric** **119**
- **CNG** **13**
- **Propane** **2,232**

Our Oldest Georgia School Buses

1989	20
1990	15
1991	1
1992	3
1993	2
1994	18
1995	34
1996	73
1997	51
1998	33
1999	153
2000	247
2001	238
2002	671
2003	587
2004	406
2005	764

2006	764
2007	627
2008	636
2009	574
2010	256
2011	703
2012	671
2013	670
2014	664
2015	924

Total number of 1989 -2015 model year units: 9525
47% of Georgia school buses

Total number of 1989 -2005
model year units: **3316**

Introducing the EnviroKlenz SMART Mount™

From real-world case studies to measurable impact, the EnviroKlenz SMART Mount™ is reshaping how transportation fleets measure and manage air quality. This first-of-its-kind solution delivers cleaner, healthier environments for passengers and drivers alike.

✓ DETECT

7 air quality conditions through real-time monitoring

✓ CORRECT

airborne risks through advanced, automated air cleaning

✓ REFLECT

on fleet health through real-time alerts & data dashboard



*What that means
for your fleet:*



Health & safety savings



Ridership & loyalty



Lower liability costs



Public recognition



Data monetization



System failure detection

Real-world impact

Explore real-world success stories: From Louisiana to Maryland, SMART Mount™ is transforming fleets with the industry's leading IAQ ecosystem.

With measurable results like a 59% reduction in PM2.5, 31.4% in TVOCs, and 44.5% in CO2, SMART Mount™ is setting the standard for cleaner, healthier transportation.





Using Propane Autogas with School Buses

Anthony V. Jackson
Executive Director, Transportation
August 13, 2024





CNG: Driving
the Past,
Present and
Future of Clean
Transportation

AGL in the Compressed Natural Gas (CNG) Value Chain

Delivery

Local Distribution Company,
Owner and Operator of the
Infrastructure



CNG Station Construction & Maintenance

“One Stop Shop” for
CNG Station Needs – we
offer station investment
solutions and
maintenance services



Transportation Sector

Great opportunities for
heavy-duty/long-haul
trucking fleets

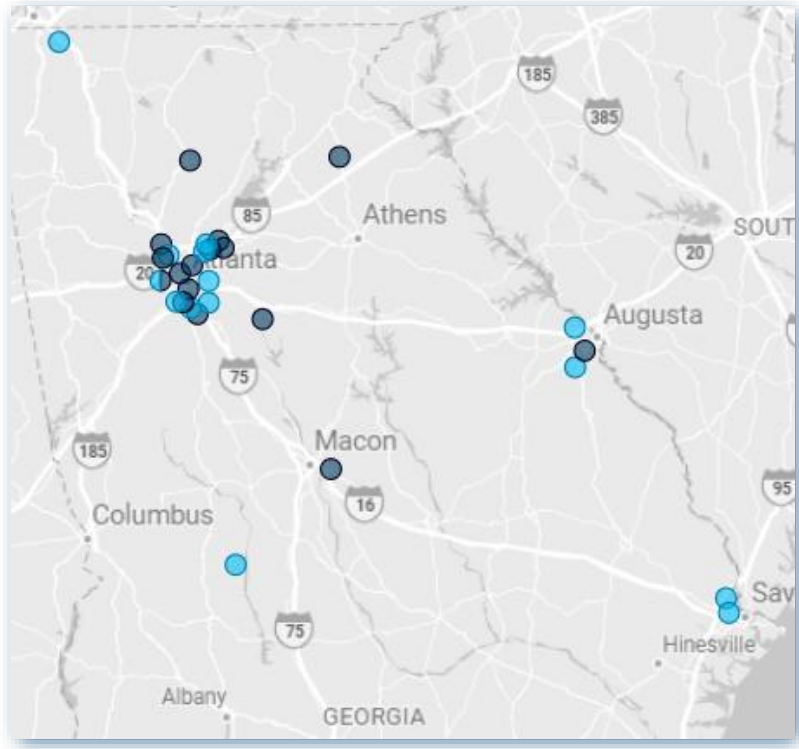


Benefits

Reduced Operating
Expenses + Scope 1
emissions reductions –
especially with RNG fuel



AGL CNG Overview



30+ years of CNG station construction, repair and maintenance experience

99% uptime for stations we do manage

On-call **24-hour emergency response** – arriving w/in an hour

Of the 54 stations in GA, **AGL owns and maintains 38** stations (including public private, customer-owned)

Industry Partners

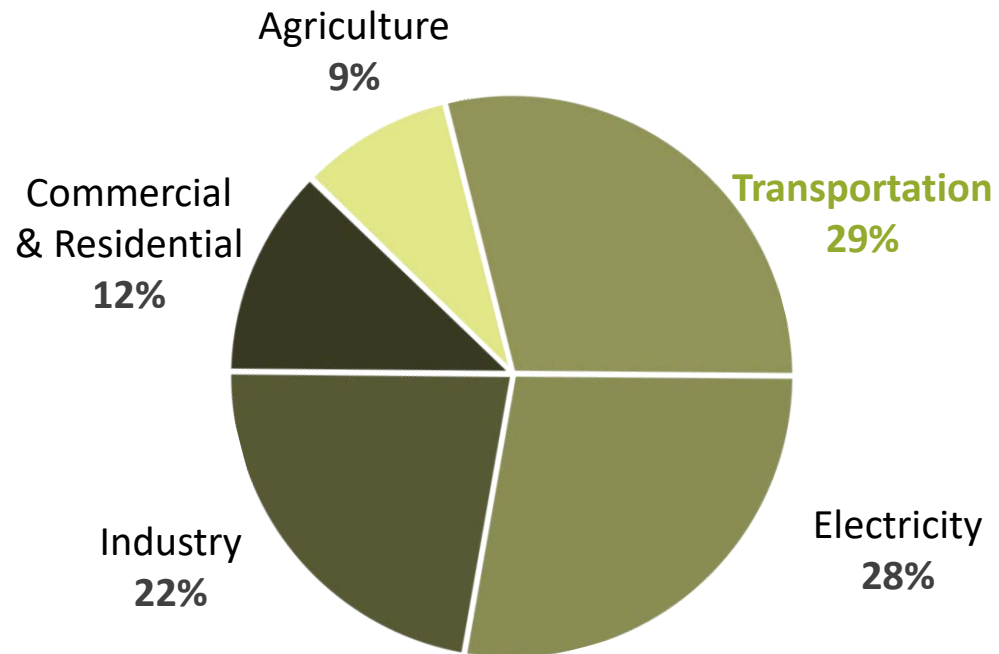


Key Customers



Taking a Closer Look At GHGs Emissions

Sources of Greenhouse Gas Emissions



29%

of all U.S. GHG emissions are
related to transportation



Of that 29%, 83% come from on-road
sources...with **25%** coming specifically
from the heavy-duty trucking sector



59%

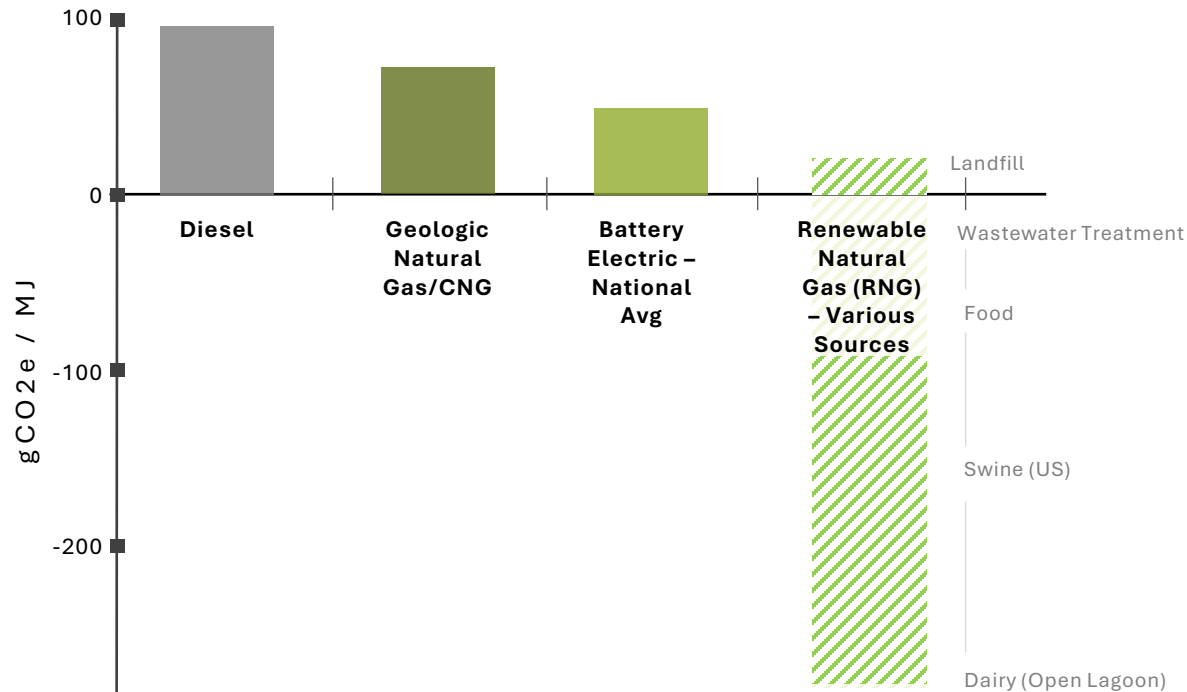


24%



1%

RNG: The Most Sustainable Transportation Fuel



Sources: EER Carbon-Intensity values based on CARB LCFS program data under CA-GREET 3.0; AgStar Project Development Handbook, 3rd Edition



CNG reduce GHGs by up to **21% compared to gasoline** or diesel.

When CNG vehicles are fueled with RNG, **“well-to-wheels” GHG emissions are drastically reduced, or even carbon negative**, depending on the RNG source.

The Southern Company Gas Renewables Team has the ability to **provide turnkey RNG solutions at stations built and/or maintained by GAS.**



Why CNG For Transportation?

Environmental Benefits



- GHG emissions are at least 20% cleaner than gasoline and diesel
- Leveraging RNG further reduces GHG emissions, with potential to be carbon-negative
- Current CNG technology meets the proposed 2027 and 2030 NOx and PM standards

Economic Benefits



- CNG truck costs are 2X-3X cheaper than emerging technology counterparts
- Natural gas commodity prices/forecasts are more stable because it is domestically-produced
- Natural gas is typically \$2 cheaper on a per gallon equivalent compared to gas/diesel

Operationally Proven



- NGVs have been fueling clean transportation for three decades
- CNG heavy-duty trucks are the only alternative fuel trucks that feature same performance characteristics and range as gasoline/diesel
- CNG heavy duty truck operators do not have to sacrifice payload or GVWR to support CNG storage equipment

Ideal Candidates – 3T's



- Trash, Transit + Trucking applications are well-suited for CNG
- Significant savings with fuel economy
- Return-to-base operations or carriers with 'daycab' or 'slip-seat' routes
- Fast-fill fueling times on par with 'fast diesel'

For More Information

Shaun Rosemond

Manager, Natural Gas Vehicles

Southern Company Gas

G2SOGASNGV@southernco.com

404.561.1012

We provide CNG Fleet Conversion Modeling and Analysis Support!



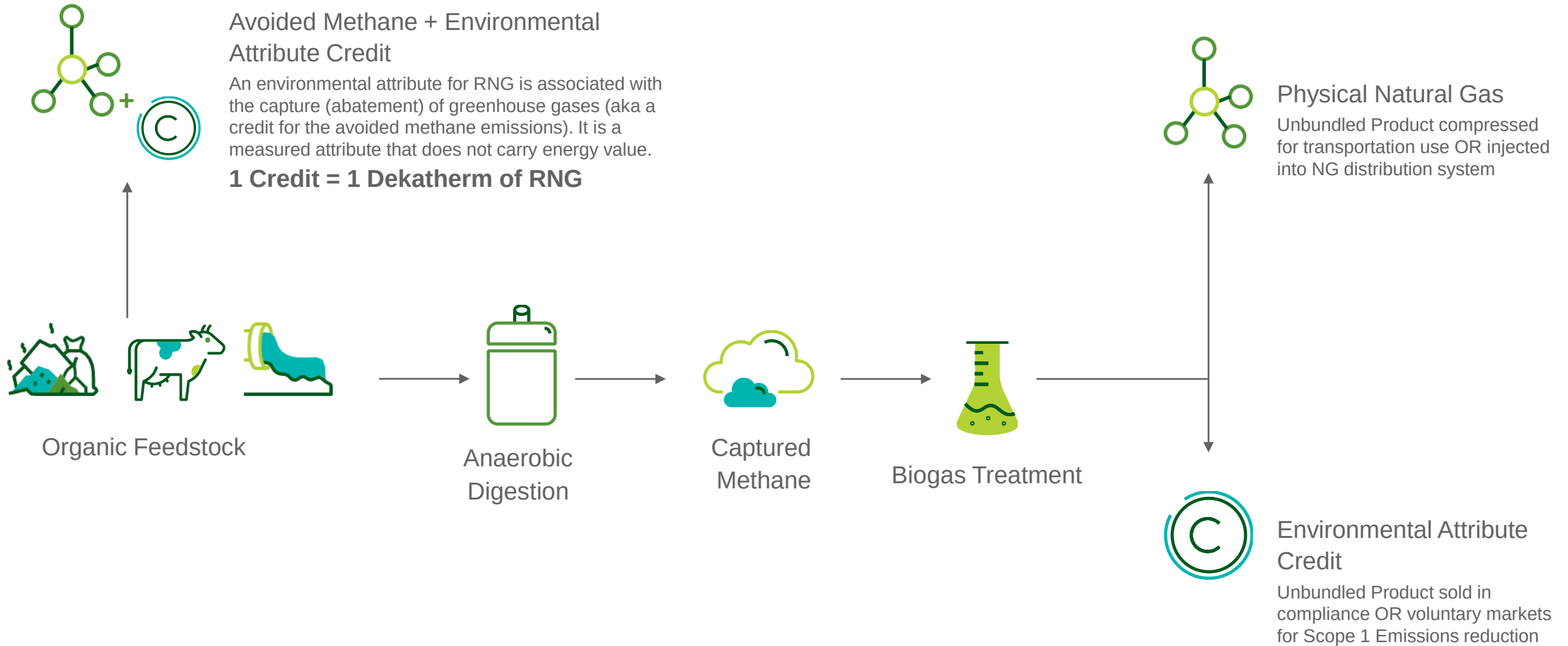
Turning Waste into Worth: RNG in Georgia

Earnell Kelly

Manager, Renewable Gas Business Development

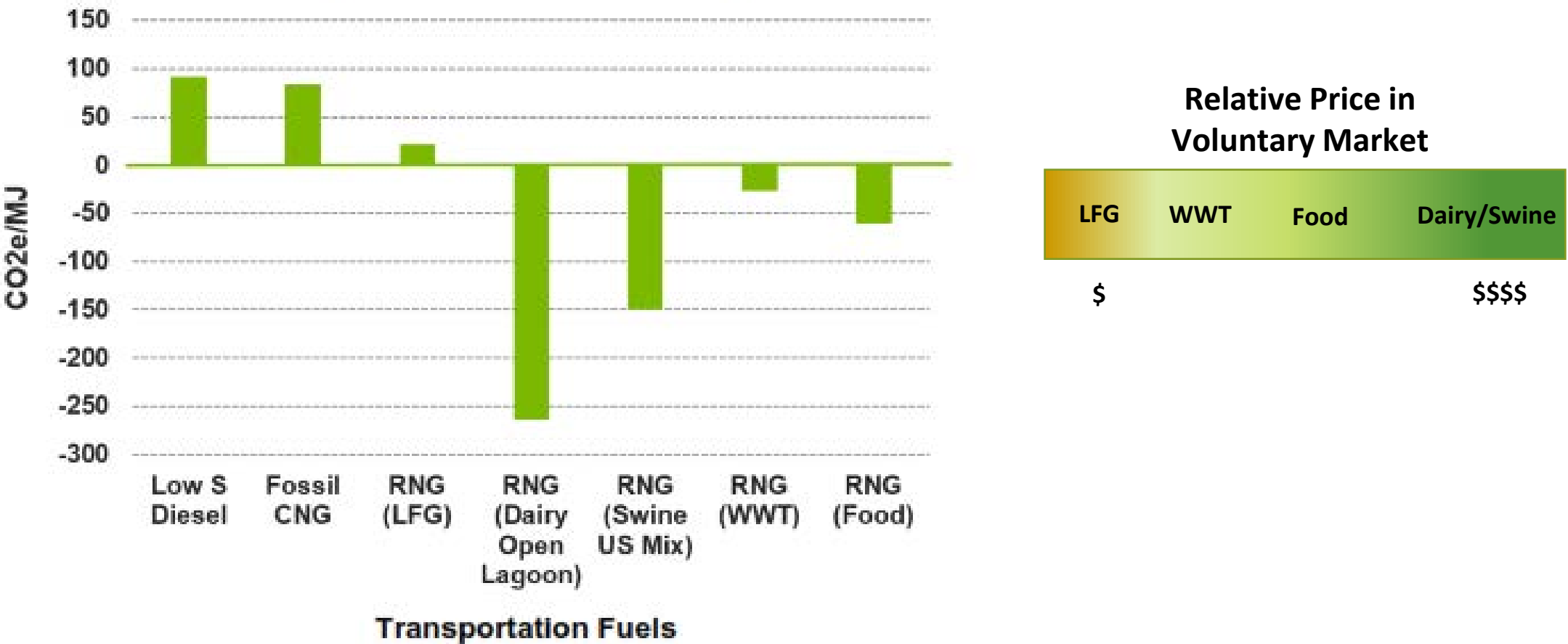


The Role of Renewable Gas



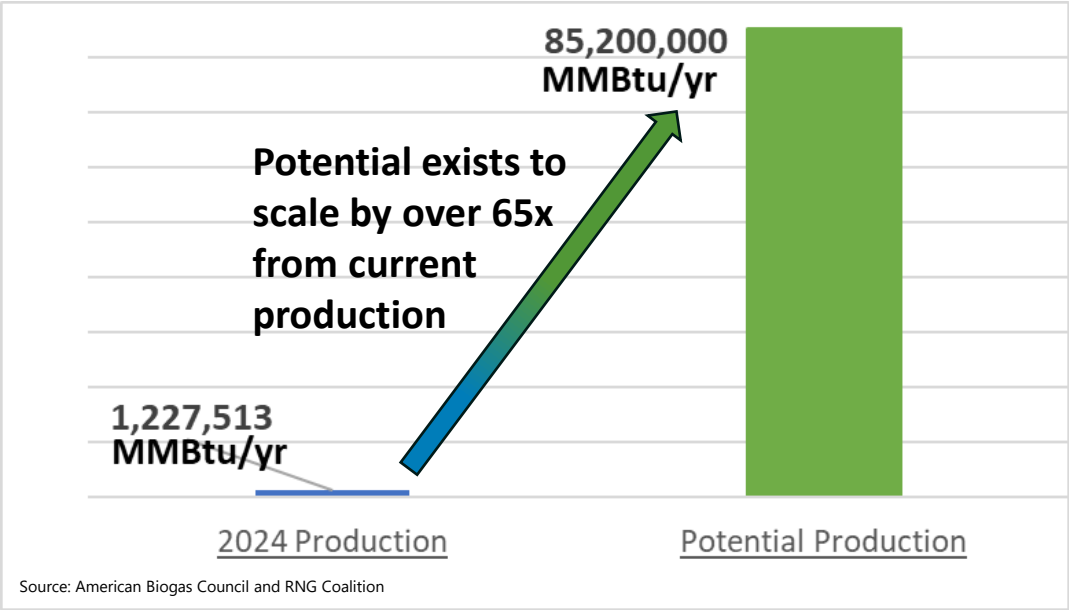
Carbon Intensity of RNG Fuels

Figure 2.8. Relative CI of Transportation Fuels – DOE Argonne National Lab¹⁸



Source: AgStar Project Development Handbook, 3rd Edition

How much RNG Potential exists in GA?



A full build-out of Georgia’s biogas industry offers these benefits:

Energy Benefits

Up to **85.2 billion ft³** of methane (renewable natural gas) could be produced each year for energy, heat, fuel, and more!

- or **11.4 billion kWh** equivalent to the annual electricity usage of **1,051,926 households**
- or **8,096 million kWh** electricity and **3,153 million BTU/h** heat (engine)
- or **13,541 million kWh** electricity and **22,568 million kWh** heat (fuel cell)
- or **1.3 GW** Nameplate Capacity equivalent to **26 U.S. power stations** (avg. size)
- or **88.2 million MMBtu/yr** equivalent to energy consumption of **1,149,067 homes**
- or **734.3 million gallons** of GGE, enough to fuel **416,048 delivery trucks** for one year

Source: American Biogas Council and RNG Coalition

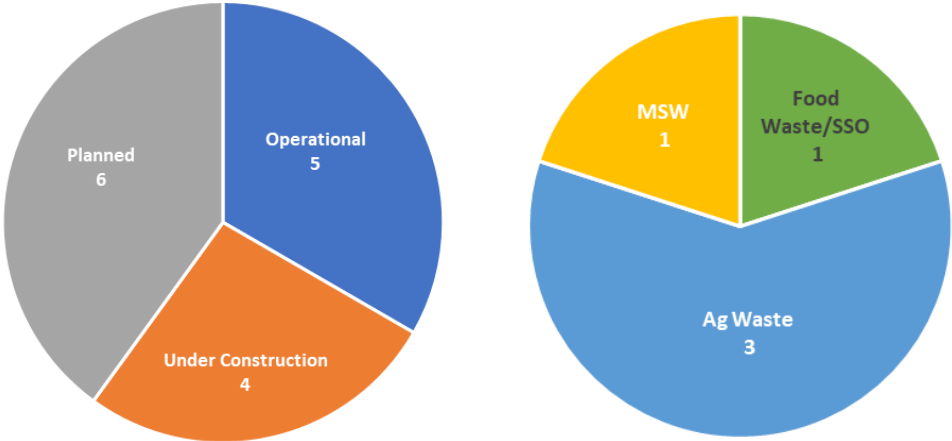


Georgia
ranks **6th** out of 50 states
for biogas production
potential of **151 bcf/yr**

Source: American Biogas Council
Georgia Biogas State Profile

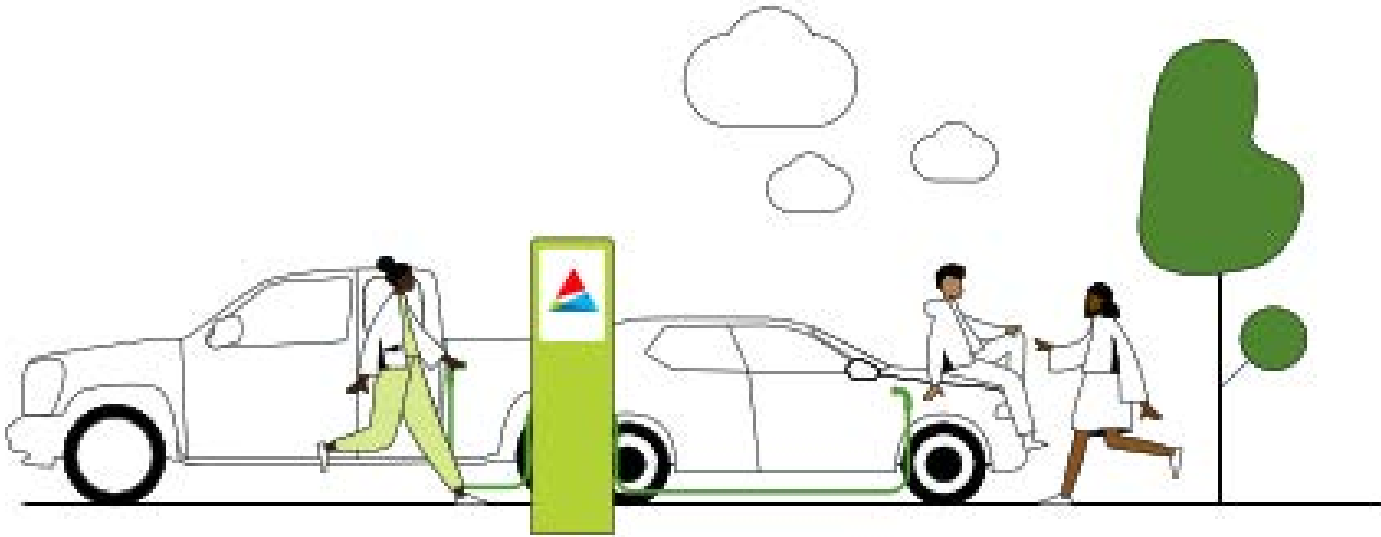
TOTAL RNG PROJECTS	GA
Operational	5
Under Construction	4
Planned	6
TOTAL	15

Source: RNG Coalition – RNG Facilities Database Master List



Source: RNG Coalition – RNG Facilities Database Master List

RNG + CNG – A Winning Combination



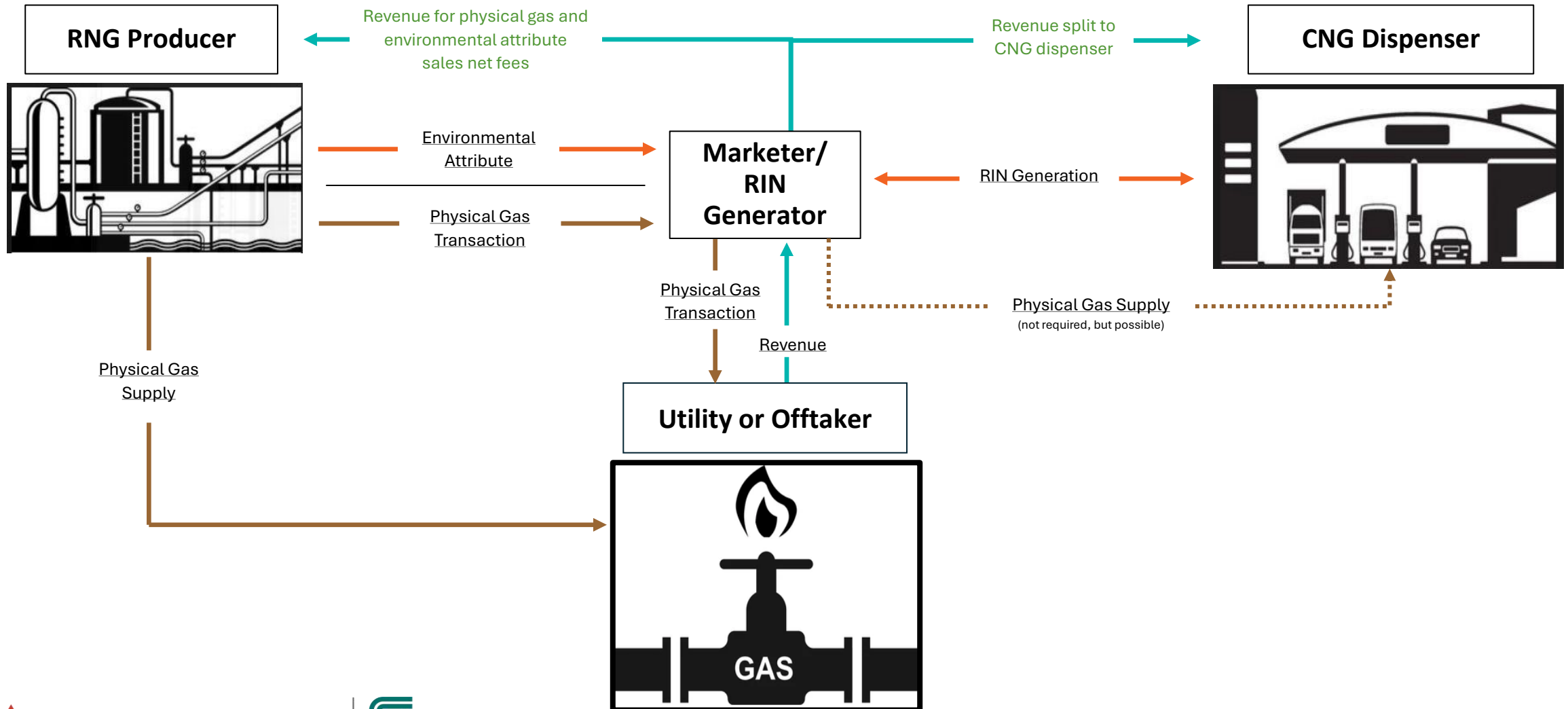
RNG-fueled trucks and buses have access to an established public fueling infrastructure and a mature network of servicers and suppliers. RNG trucks are affordable and scalable, meeting range and operations requirements for any commercial medium- and heavy-duty application. Also, RNG fueled trucks and buses do not rely on components sourced and controlled overseas^[2].

Did you know?

RNG reduces 660 million gallons of diesel consumed by heavy duty vehicles. By filling approximately 3 million semi-trucks or 7.3 million transit buses, 14,792 million pounds of CO₂ emissions can be reduced by converting existing waste streams into a valuable fuel source^[1].

In 2023, 79 percent of all natural gas used in transportation in the U.S. was RNG. The industry aspires to reach 80 percent by 2030 and virtually 100 percent by 2050^[2].

Typical RNG Dispensing Agreement



Thank you

An aerial photograph of a large, single-story commercial building with a brown roof and light-colored siding. The building has a sign that reads "DeKalb County Public Works Fleet Management" and the address "5350". In the foreground, a large white dump truck is parked in a parking lot. To its right, a white garbage truck with a yellow compactor is parked. Several other vehicles, including a silver pickup truck, a white van, and a dark car, are parked nearby. The parking lot is paved and has some trees and shrubs around the building.

Driving Sustainability with Alternative Fuels

DeKalb County, Fleet Management

Deputy Director, Robert Gordon



DeKalb County
GEORGIA

Cleaner Miles Ahead: Our Alternative Fuel Vehicles

Fleet has deployed over 525 alternative fuel vehicles, which represent 14% of our total fleet of 3,678 vehicles and equipment.

Fuel Type	Total Vehicles
CNG	349
Electric	114
Propane	35
Hybrid	25
Bi-Fuel	2

CNG Powered Vehicle Types

Heavy Duty/ Medium Duty

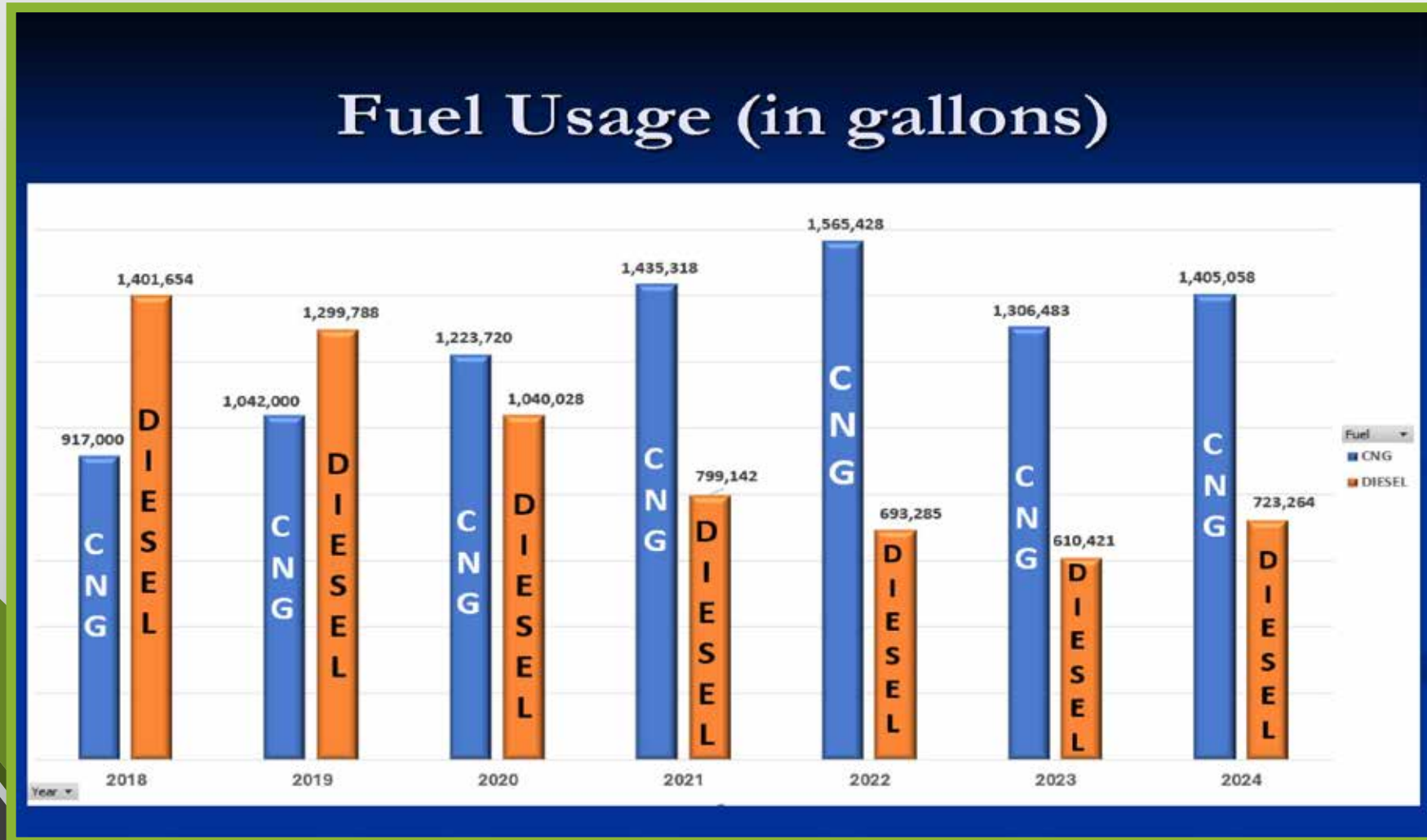
- Rear Loaders
- Front Loaders
- Side Loaders
- Automated Side Loaders
- Roll Off Trucks
- Grappler Trucks
- Crew Trucks
- Crane Trucks
- Level Lift Trucks
- Bucket Truck
- Road Tractors
- Road Fork Trucks

Light Duty

- Pickup Trucks
- Sedans
- Vans
- Side Loaders
- Missed Collections Trucks



Fleet Fuel Consumption Report



Charging Ahead: What's Next for Fleet

- Fleet currently operates 72 Level 2 charging stations.
- Coming September 2025:
 - Addition of 46 Level 2 chargers.
 - Installation of 2 Level 3 fast charging ports.
 - Multiple locations will be accessible for public use.
- Planned for Q1 2026:
 - Installation of 50 additional Level 2 charging stations.
 - 40 of these stations will be shared with the public.



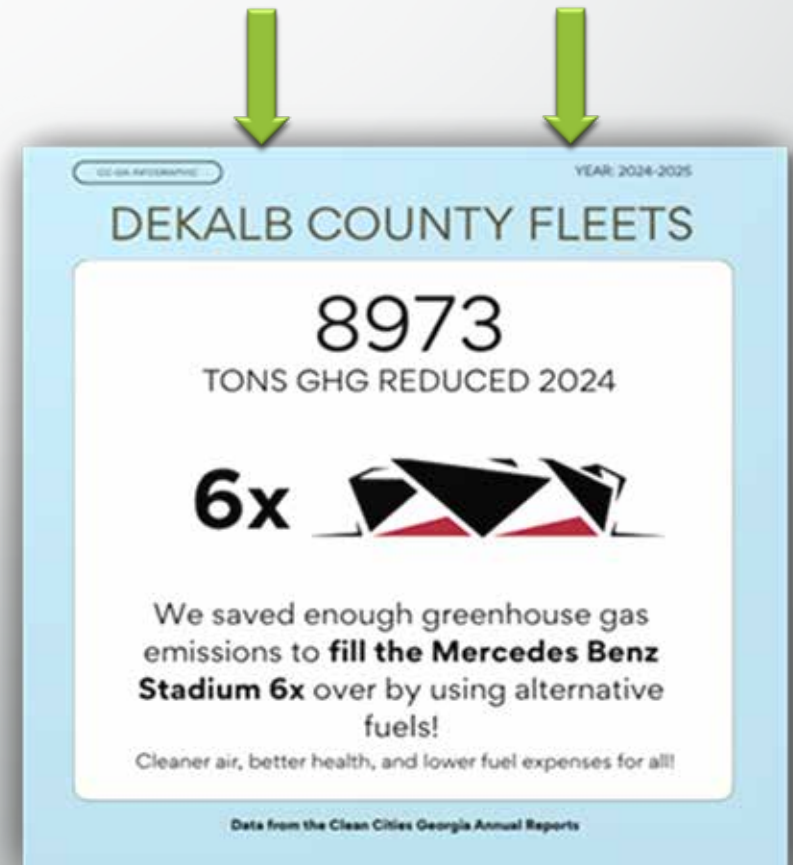
For More Information

SCAN THE QR CODE BELOW TO WATCH THE DEPUTY DIRECTOR, ROBERT GORDON DIVE INTO ALL OF FLEET'S ALTERNATIVE FUEL VEHICLES.



Skip to 7:26

CHECK OUR STATS



Let's Stay Connected

Robert Gordon
Deputy Director, Fleet Management



rlgordon1@dekalbcountyga.gov



(404) 297-3280



(770) 652-6972



[Welcome to Fleet Management](#)
[| DeKalb County GA](#)









Caleb Kitchings

Director of Business Development
Savannah Economic Development Authority

Moderator



Joshua Vidic

Senior Group Manager
Hyundai Motor America

Panelist



Caitlin Rackish

Senior Project Manager
HydroFleet

Panelist



Hydrogen: Why Hydrogen?

Zero Emission

When utilized in fuel cells, hydrogen generates electricity, with only water vapor and heat as byproducts. This renders it a clean and environmentally friendly energy source, particularly in contrast to high carbon emitters that release greenhouse gases, contributing to environmental degradation.

Versatility

Hydrogen can be used in various sectors, including transportation, industry, and power generation. It can be used as a fuel for fuel cell vehicles, blended with natural gas for heating, or used in industrial processes such as refining and manufacturing.

Storage

Hydrogen can serve as a means of energy storage. Excess energy generated from renewable sources, such as wind or solar power, can be used to produce hydrogen through electrolysis. This hydrogen can then be stored and later used to generate electricity or power vehicles when renewable energy generation is low.

High Density

Hydrogen has a high energy density, which means it can store a large amount of energy in a small volume or mass. This makes it a potentially efficient energy carrier for applications where space and weight are critical, such as in transportation.

Minimal Output

Hydrogen provides an alternative to traditional high carbon-emitting energy sources, helping to diversify the energy mix and reduce dependence on non-renewable resources. This can contribute to energy security and mitigate the environmental impact of high carbon output from energy consumption.

Hydrogen: Application



Fuel Cells: Converts chemical energy from hydrogen into electricity, powering everything from vehicles to buildings, with water as the only emission.



Combustion Engines: Hydrogen mixes with air and ignites like traditional fuels, producing high energy output and water vapor as the primary emission, offering a cleaner alternative to gasoline or diesel.



Industrial Processes: Used in refining, treating metals, and producing fertilizers.



Data Centers: Use Hydrogen fuel cells to provide clean, reliable backup power, enhancing energy efficiency compared to traditional diesel generators.



Energy Storage: Acts as a buffer for renewable energy sources, storing excess energy and delivering it back when needed.



Portable Power: Convert hydrogen into electricity for on-the-go energy solutions, emitting only water. (US Army)

Hydrogen Safety Considerations



Emergency Response

Ensure adequate emergency response procedures and equipment are available to end users. Maintain firefighting equipment/infrastructure nearby hydrogen installations/facilities. Educate first responders on hydrogen-related emergency response.



Explosion

Explosion may result from the accumulation of hydrogen gas in an enclosed space. Ensure proper ventilation in areas where flammable hydrogen gas may accumulate. Implement emergency shut-off devices capable of remote operation.



Flame Jetting

Install flame detectors, considering the poor visibility and low radiant heat of hydrogen flames. Ensure proper system design and shutdown planning to mitigate the risks associated with flame jetting. Provide passive fire protection measures and limit affected areas in the event of a fire.



Ignition Prevention

Hydrogen has a broad flammability range and is very easy to ignite. Anticipate likely sources of hydrogen leaks and implement measures to reduce risk. Avoid potential ignition sources near hydrogen-servicing equipment. Use grounded/bonded non-sparking equipment to minimize the risk of ignition and fire.



Leak Detection

Hydrogen is odorless and colorless, making it hard to detect with the human senses. Install sensors to detect hydrogen releases. Sensors should be interlocked to shut off the hydrogen supply and/or actuate alarms.



Loss of Hydrogen Containment

Leaking hydrogen may lead to flame jetting and/or explosion. To mitigate this risk, utilize high-quality materials that meet standards and are compatible with hydrogen to prevent material defects and leaks. Follow material requirements outlined in the relevant codes and standards to minimize embrittlement and corrosion risks. Enhance the integrity and reliability of hydrogen systems by using welded connections wherever possible.



Oxygen Displacement

Maintain adequate ventilation to prevent the displacement of oxygen with hydrogen in enclosed spaces, which may lead to asphyxiation.

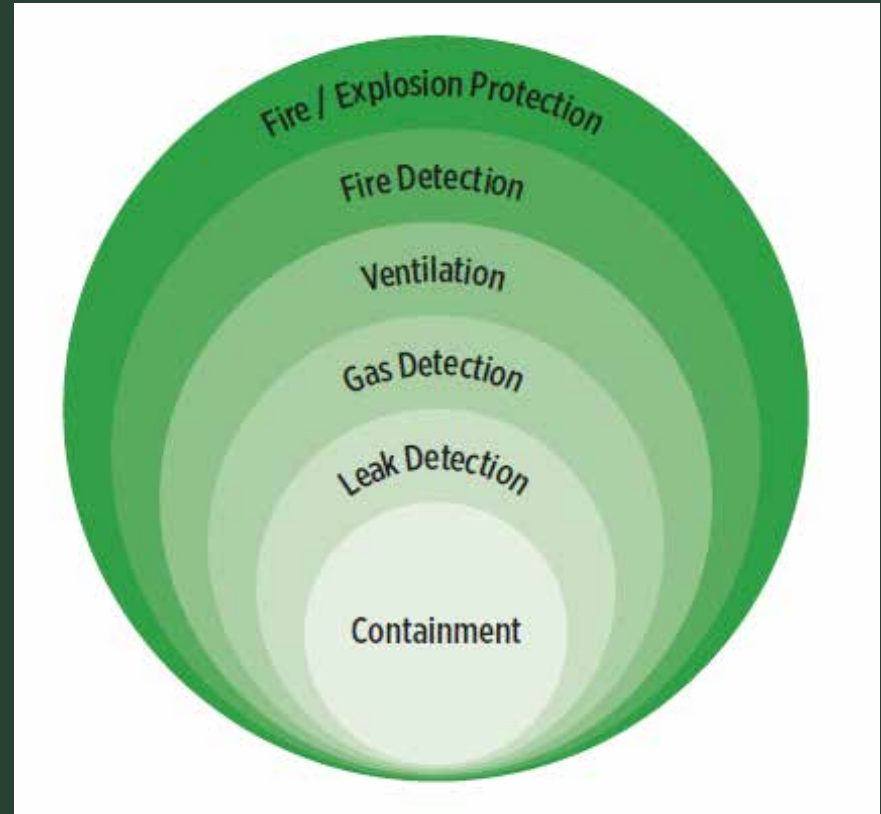


Pressure and Temperature

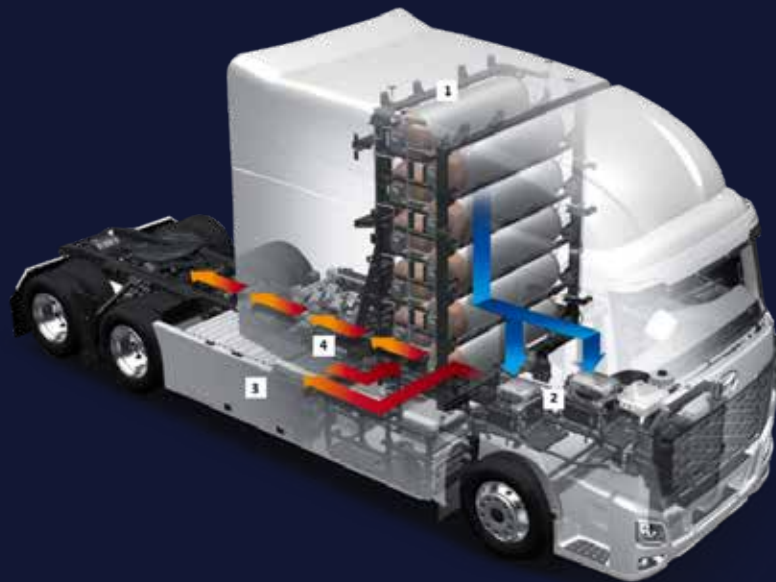
Overpressure may result in loss of containment and/or kinetic damage from equipment perforation. High temperatures may exacerbate this risk and present thermal contact hazards. Cryogenic liquid hydrogen presents thermal hazards as well. Incorporate appropriate pressure relief devices in high-pressure gas systems. Establish proper handling procedures and safety protocols for cryogenic liquid hydrogen. Use sensors, relief valves, control and alarm instrumentation, and regular maintenance protocols for temperature and pressure control.

H2 Transport and Storage Safety System

To ensure safe operation companies producing, transporting, storing and using hydrogen will use a multi-layered safety system. The first layer is containment in secure pipeline or distribution and storage technologies. The next layers are leak and gas detection. Ultrasonic detectors can identify minute leaks and catalytic and electrochemical detectors can sense discrete amounts of hydrogen. Ventilation can prevent hydrogen leak ignition, but recognize that hydrogen has an extremely low ignition point. Fire/flame detection, monitoring infrared and ultraviolet radiation can detect colorless/odorless hydrogen combustion and initiate automated fire suppressant systems. Finally, fire and explosion protection through facility location and material use can protect workers and other assets in the event of an incident.



Hyundai XCIENT FCEV Class 8 Tractor



HIGH-TECH HYDROGEN SYSTEM



(1) Hydrogen Tank:
68.6 kg H₂ (700 bar)



(2) Stack:
180 kW (2 x 90 kW each)



(3) Battery:
72 kWh (3 x 24 kWh each)



(4) Motor:
350 kW / 1,650 lbf-ft



US



Max power output
350kW



Fuel Cell system
2x90kW



Battery
72kWh



Hydrogen capacity
68kg/700bar



Target Range
Over **450miles**



Refueling Time
Around **30min.**

HTWO Energy Savannah



HTWO Energy Savannah





SHEA

Southeast Hydrogen Energy Alliance

GA Clean Energy Roadshow
August 13-14, 2025





Dr. Scott McWhorter

Georgia Tech, Strategic Energy Institute

Lead, Federal Energy Initiatives

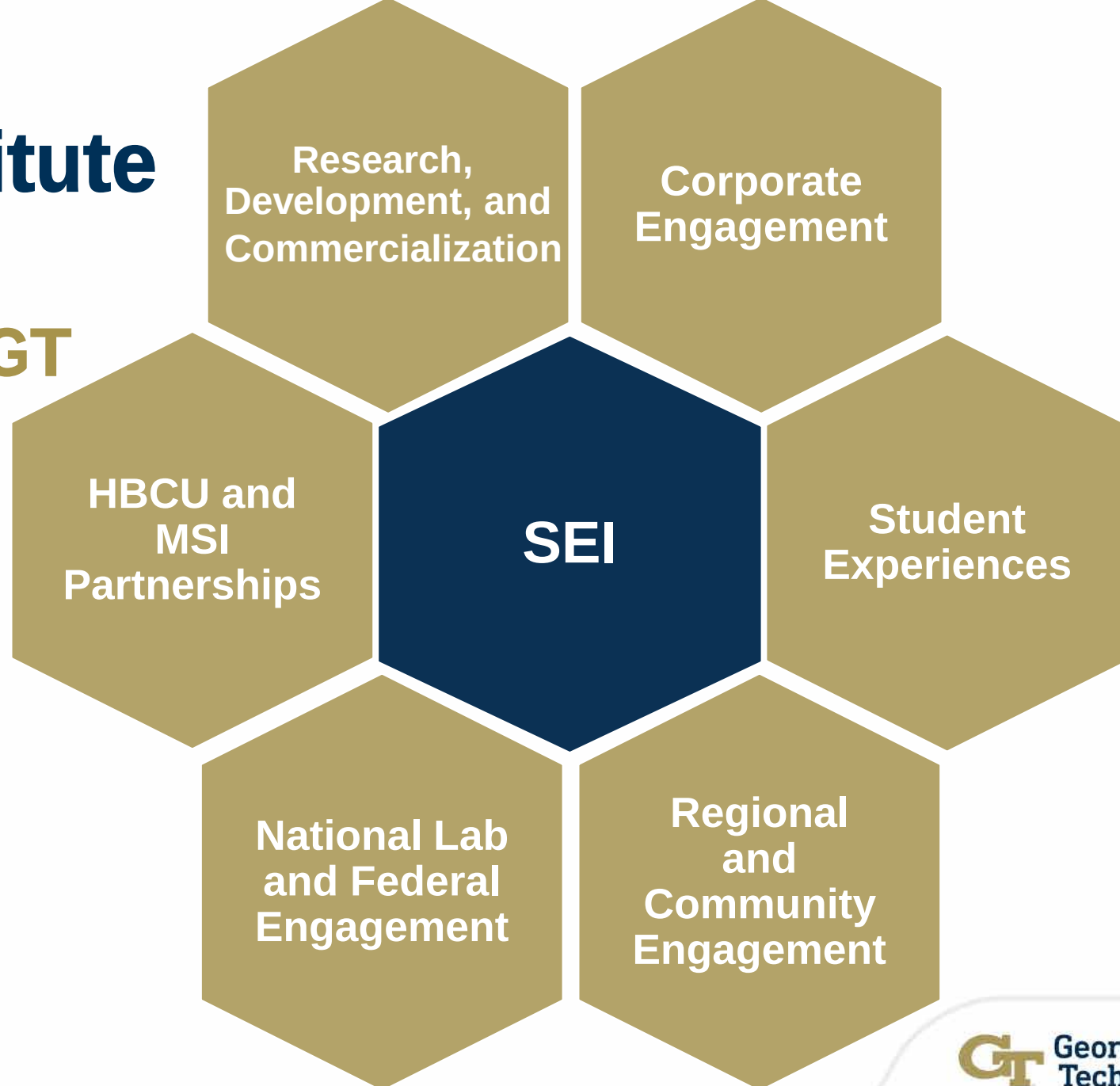
Board Chairman, Southeast Hydrogen Energy Alliance

- **Founder, FuelX Innovations**, developing advanced hydrogen storage systems.
- **Consultant** in federal business development and technology start-ups
- **Former Division Director of Clean Energy Research** at Savannah River National Laboratory (SRNL), securing over **\$30 million** in new funding across DOE programs (Office of Science, EERE, OE, ARPA-E, and FE).
- **21-year career at SRNL**, including 3 years as **Technical Advisor** with the U.S. DOE EERE Hydrogen Fuel Cell Technologies Office in Washington, DC.



Strategic Energy Institute

- **1 of 9 Interdisciplinary Research Institutes at GT**
- **Amplifying Impact, Trusted Partners**



Tools to Learn & Grow



www.seh2.energy



Login to your Membership Portal
to unlock all the potential



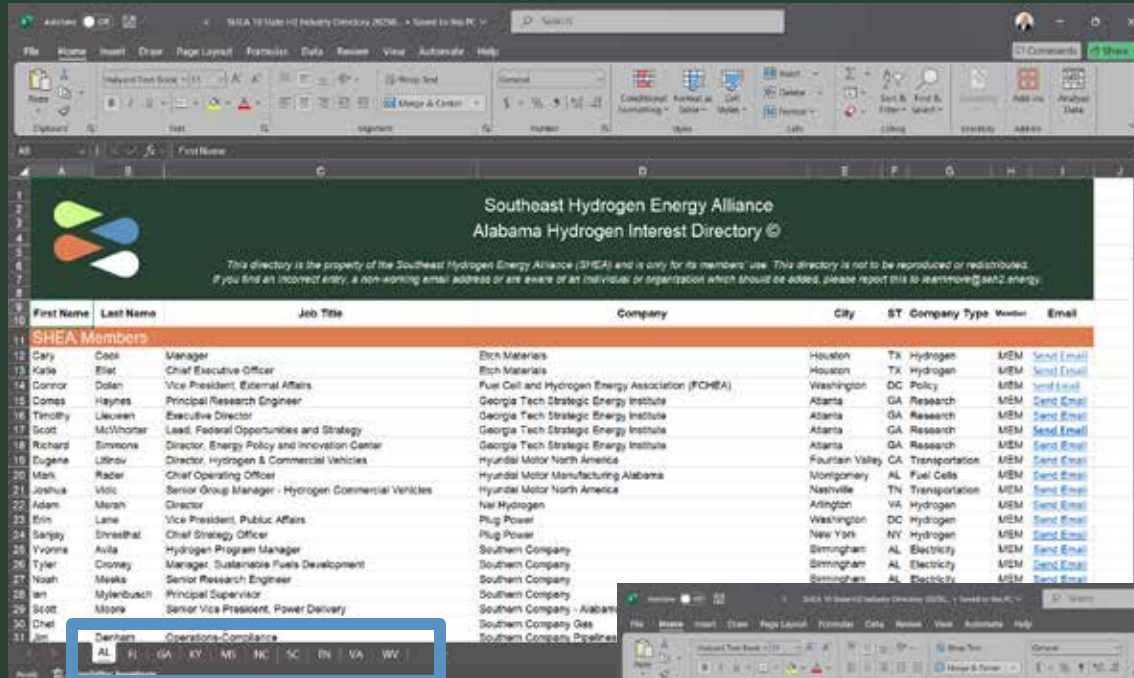
Built by Members, Driven by Impact

From utilities and OEMs to state agencies and startups, this community is the engine behind every tool, connection, and initiative SHEA brings to life.



Member Resource

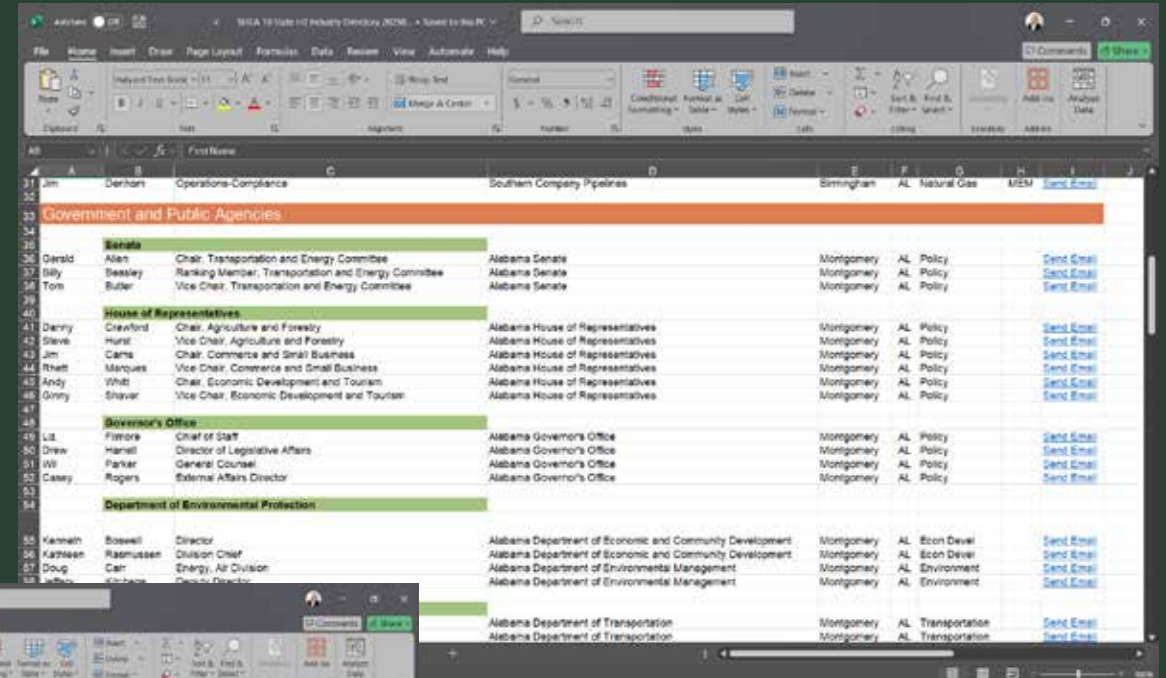
10 State Hydrogen Database Preview



Southeast Hydrogen Energy Alliance
Alabama Hydrogen Interest Directory ©

This directory is the property of the Southeast Hydrogen Energy Alliance (SHEA) and is only for its members' use. This directory is not to be reproduced or redistributed. If you find an incorrect entry, a non-working email address or are aware of an individual or organization which should be added, please report this to jeanm@shea2.energy.

First Name	Last Name	Job Title	Company	City	ST	Company Type	Member	Email
SHEA Members								
11	Cary	Cook	Manager	Etch Materials	Houston	TX Hydrogen	MEM	Send Email
12	Kate	Eliot	Chief Executive Officer	Etch Materials	Houston	TX Hydrogen	MEM	Send Email
13	Connor	Dolan	Vice President, External Affairs	Fuel Cell and Hydrogen Energy Association (FCHEA)	Washington	DC Policy	MEM	Send Email
14	Connes	Haynes	Principal Research Engineer	Georgia Tech Strategic Energy Institute	Atlanta	GA Research	MEM	Send Email
15	Timothy	Leuven	Executive Director	Georgia Tech Strategic Energy Institute	Atlanta	GA Research	MEM	Send Email
16	Scott	McWhorter	Lead, Federal Opportunities and Strategy	Georgia Tech Strategic Energy Institute	Atlanta	GA Research	MEM	Send Email
17	Richard	Simmons	Director, Energy Policy and Innovation Center	Georgia Tech Strategic Energy Institute	Atlanta	GA Research	MEM	Send Email
18	Eugene	Litvin	Director, Hydrogen & Commercial Vehicles	Hyundai Motor North America	Fort Worth	TX Hydrogen	MEM	Send Email
19	Mark	Rader	Chief Operating Officer	Hyundai Motor Manufacturing Alabama	Montgomery	AL Hydrogen	MEM	Send Email
20	Joshua	Wolfe	Senior Group Manager - Hydrogen Commercial Vehicles	Hyundai Motor North America	Nashville	TN Hydrogen	MEM	Send Email
21	Adam	Marsh	Director	Nal Hydrogen	Arlington	VA Hydrogen	MEM	Send Email
22	Erin	Lane	Vice President, Public Affairs	Plug Power	Washington	DC Hydrogen	MEM	Send Email
23	Sanjay	Shrestha	Chief Strategy Officer	Plug Power	New York	NY Hydrogen	MEM	Send Email
24	Yvonne	Avila	Hydrogen Program Manager	Southern Company	Birmingham	AL Hydrogen	MEM	Send Email
25	Tyler	Cronay	Manager, Sustainable Fuels Development	Southern Company	Birmingham	AL Hydrogen	MEM	Send Email
26	Noah	Meeks	Senior Research Engineer	Southern Company	Birmingham	AL Hydrogen	MEM	Send Email
27	Mylenbuch	Moore	Principal Supervisor	Southern Company	Birmingham	AL Hydrogen	MEM	Send Email
28	Scott	Moore	Senior Vice President, Power Delivery	Southern Company Gas	Birmingham	AL Hydrogen	MEM	Send Email
29	Jim	Denham	Operations-Compliance	Southern Company Pipelines	Birmingham	AL Hydrogen	MEM	Send Email



Government and Public Agencies

Senate								
31	Gerold	Allen	Chair, Transportation and Energy Committee	Alabama Senate	Montgomery	AL Policy	Send Email	
32	Billy	Beasley	Ranking Member, Transportation and Energy Committee	Alabama Senate	Montgomery	AL Policy	Send Email	
33	Tom	Bulder	Vice Chair, Transportation and Energy Committee	Alabama Senate	Montgomery	AL Policy	Send Email	
House of Representatives								
34	Denny	Crawford	Chair, Agriculture and Forestry	Alabama House of Representatives	Montgomery	AL Policy	Send Email	
35	Steve	Hurst	Vice Chair, Agriculture and Forestry	Alabama House of Representatives	Montgomery	AL Policy	Send Email	
36	Jim	Carr	Chair, Commerce and Small Business	Alabama House of Representatives	Montgomery	AL Policy	Send Email	
37	Rhett	Marques	Vice Chair, Commerce and Small Business	Alabama House of Representatives	Montgomery	AL Policy	Send Email	
38	Andy	White	Chair, Economic Development and Tourism	Alabama House of Representatives	Montgomery	AL Policy	Send Email	
39	Ginny	Shaver	Vice Chair, Economic Development and Tourism	Alabama House of Representatives	Montgomery	AL Policy	Send Email	
Governor's Office								
40	Liz	Finnore	Chief of Staff	Alabama Governor's Office	Montgomery	AL Policy	Send Email	
41	Drake	Harrell	Director of Legislative Affairs	Alabama Governor's Office	Montgomery	AL Policy	Send Email	
42	Walt	Parham	General Counsel	Alabama Governor's Office	Montgomery	AL Policy	Send Email	
43	Casey	Rogers	External Affairs Director	Alabama Governor's Office	Montgomery	AL Policy	Send Email	
Department of Environmental Protection								
44	Kenneth	Boswell	Director	Alabama Department of Economic and Community Development	Montgomery	AL Policy	Send Email	
45	Kathleen	Rasmussen	Division Chief	Alabama Department of Economic and Community Development	Montgomery	AL Policy	Send Email	
46	Doug	Carroll	Energy, Air Division	Alabama Department of Environmental Management	Montgomery	AL Policy	Send Email	
47	Jeffrey	Ward	Deputy Director	Alabama Department of Environmental Management	Montgomery	AL Policy	Send Email	
Alabama Department of Transportation								
48	Jeffrey	Ward	Deputy Director	Alabama Department of Transportation	Montgomery	AL Policy	Send Email	

All 10 states represented so easily searchable, but protecting the emails from mass downloads

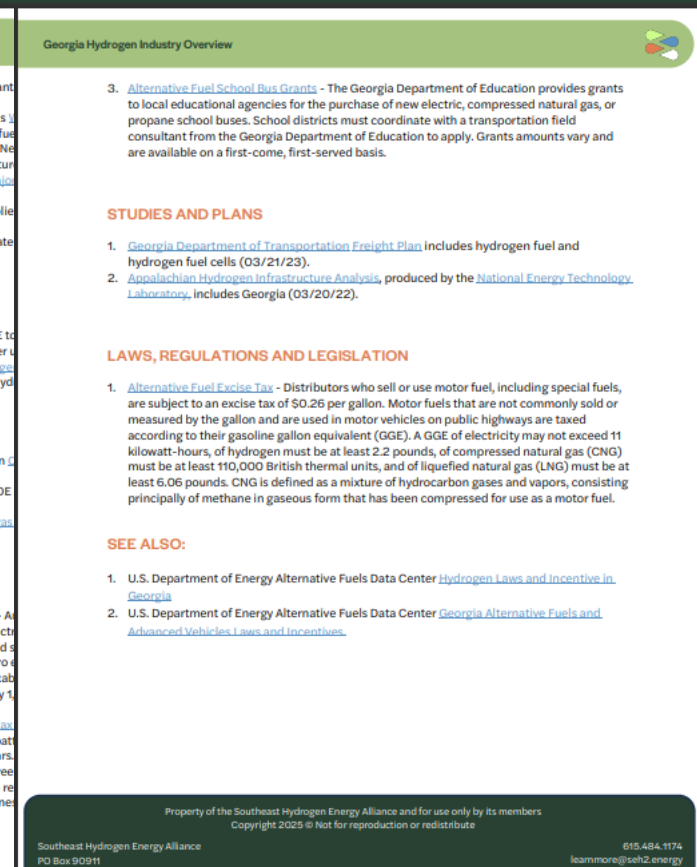
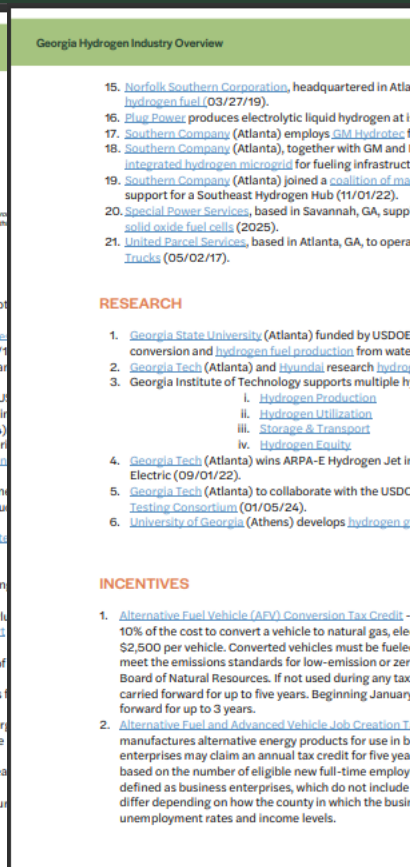
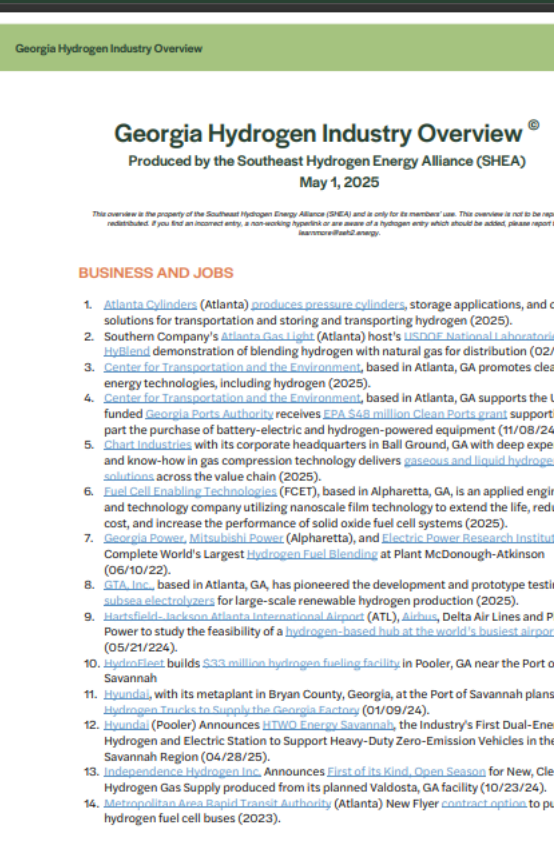
Sections Include by State:

1. Members
2. Government & Public Agencies
3. Non-Profit Organizations
4. University & Research Organizations
5. Business & Industry



Member Resource

SHEA State Summary Packet



[Member Resources - SHEA](#)

SuperTruck 3 with General Motors + Southern Company

Demonstrate 75% GHG reduction with TCO parity with combustion engines

Argonne
NATIONAL LABORATORY

Simulation and Modeling
of Adoption & Impact



Southern Company
Scalable H2
Infrastructure via
Electric Grid Coupling



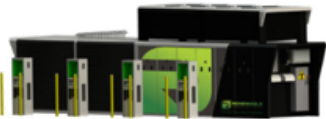
Electric Grid

Microgrid
Controller



Photovoltaics

Stationary Fuel Cell
Power Generation



DCFC



Battery Electric MD Trucks

nel



Prototype
Electrolyzers

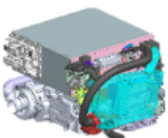


H2
Storage



H2
Dispensing

Adv NG FCS



Fuel Cell MD Trucks



Southern Company
Utility Fleet
Operator

Pending
Freight Fleet
Operator

Power
Hydrogen

Demonstration in GA



*Images are intended to be a representation, not depict the final hardware design

Georgia National Electric Vehicle Infrastructure (NEVI) Deployment Program

GDOT Roadshow
Christina Barry, PE
August 14, 2025



Scan to visit
Georgia NEVI
Website





NEVI Program

Purpose



Create a convenient and reliable nationwide network of 500,000 EV chargers by 2030

Goals



Accelerate equitable adoption of EVs, reduce greenhouse gas emissions, and encourage U.S. industries to lead global electrification

Funding



Funding rescinded per Federal Guidance; GA Round 2 put on hold

NEVI Plan



NEVI requires all states to prepare a plan compliant with federal requirements from Federal Highway Administration and the Joint Office of Energy and Transportation

**Plan Approval
Rescinded 2/7/25**
**Updated Guidance
Issued 8/11**

Georgia NEVI Deployment

- Deployment plan identified Gaps in AFC Network that are required to be addressed (noted in yellow on the map)
- Round 1 Locations represent a radius around the exit/interchange where a charging station could be located to build out the Gap
- Contracts for Round 1 Locations awarded and moving into Construction



NEVI Round 1 Deployment – Site Comparison Table

PI Number	Location	Developer	Project Status	Services Commencement
0019828	Tifton Waffle House	EnviroSpark	Design accepted; pending NTP - Construction	Q4 2025
0019829	Ft. Valley Wendys	Francis Energy	NTP – Construction signed 8/11	Q4 2025
0019830	Dublin RaceTrac	Silver Comet Energy	Design accepted; pending NTP - Construction	Q4 2025
0019831	Metter Shell	EnviroSpark	Coord. with HSO for equipment install ongoing	Q2 / Q3 2026
0019832	Brunswick Pilot Truck Care Center	Love's Travel Stops & Country Stores, Inc.	Design Plan Review ongoing	Q1 / Q2 2026

Round 1 GA NEVI Locations

Georgia NEVI Challenges



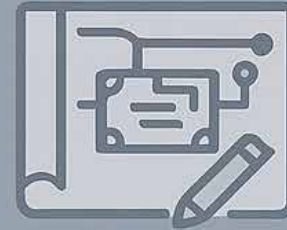
Funding Freeze

A federal
funding pause
began on
February 7, 2025



Cybersecurity

New federal
requirements
for EVSE
hardware
and software



Host Site Coordination

Misalignment
on site design,
access and
utility
readiness



Environmental Reviews

Re-
certification
needed for
several sites

FHWA Issued Updated NEVI Program Guidance on 8/11

Stay Connected



@GeorgiaDOT



@GADeptofTrans



@gadeptoftrans



www.dot.ga.gov

Thank You

jhibbard@dot.ga.gov

Phone: (404) 631-1401

Email: [Jhibbard@dot.ga.gov](mailto:jhibbard@dot.ga.gov)



CITY OF SAVANNAH

Office of Sustainability

Planning and Economic Development Service Center

100% SAVANNAH CLEAN ENERGY PLAN



PEOPLE • PLANET • PROSPERITY

Key Focus Area: Transportation and Mobility

“To meet all energy needs with renewable energy by 2050, we will transition to modes of transportation that do not require fossil fuels such as electric cars, buses, and trolleys, as well as increased use of bicycles and walking.”

City EV Portfolio

36 Light -Duty EVs

50 + EV Chargers

